Envelope Equivariant Graph Reinforcement Learning and Neighborhood Proximal Allocation for Cooperative Deadlock Resolution of Multi-Robot Systems Under Coupled Safety Constraints

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***Abstract*—Multi-robot systems operating under mutually conflicting global connectivity maintenance and pairwise collision-avoidance scalar constraints frequently suffer from non-target geometric deadlocks in dense symmetric encounters, channel blockage by outer stationary target-holding neighbors, and terminal steady-state hovering induced by learned zero-order repulsive residuals. This paper proposes an Envelope Equivariant Graph Policy with Neighborhood Proximal Allocation (EGP-NPA) cooperative control framework. First, we construct an exact rational second-order Karush–Kuhn–Tucker (KKT) certificate on a seven-robot symmetric contact configuration, proving algebraically that even when all robots are freely mobile and an explicit collision-free, connected path exists, any myopic Control Barrier Function (CBF) safety filter enforcing monotone task-error descent ($\dot{W} \leq 0$) is trapped by a positive-definite Lagrangian Hessian over the entire critical cone and a finite-radius quadratic energy barrier. This certificate identifies two key physical mechanisms for deadlock resolution: lateral symmetry-breaking nominal drive from swapping robots and dynamic quota-driven yielding from outer stationary neighbors. Second, we construct a planar $\mathrm{SO}(2)$ rotation-, translation-, and permutation-equivariant message-passing Graph Neural Network (GNN) residual nominal policy, proving the structural necessity of chiral rotational bias over full $\mathrm{O}(2)$ reflection equivariance. Coupled with a closed-neighborhood block NPA safety shield with antisymmetric edge-ledger local settlement, we derive the explicit positive closed-form solution for pairwise quota transfer and stationary neighbor yielding velocity on contact boundaries, and show how penalizing normal filter intervention weakens cooperative yielding. Furthermore, a stateless one-hop closed-neighborhood completion envelope is proposed to suppress zero-order pairwise repulsive biases to $\mathcal{O}(\|e_k\|_2)$ near the goal and restore the zero-velocity target equilibrium. Finally, trained via Multi-Agent Proximal Policy Optimization (MAPPO) on small teams ($N \in \{6, \ldots, 10\}, B = 2$) and deployed zero-shot to larger teams ($N \in \{12, 20, 25\}, B = 8$), EGP-NPA achieves 100% safe arrival and zero safety violations across 96 dense triangular-lattice swap/permutation benchmarks and 48 independent holdout instances, outperforming shield-consistent safe MARL (53.1%), static-split CBF shielded GNN (3.1%), task-adapted Discrete GCBF Proximal Policy Optimization (DGPPO, 0% violation-free arrival), primal-dual MAPPO-Lagrangian (0% violation-free arrival), tangential circulation heuristics (40.6%), and myopic CBF filtering (15.6%). Matched-budget factorial tests, fixed-weight ablations, and second-order `MuJoCo` simulations further validate each module and characterize the safety-margin versus channel-passability trade-off.**

***Index Terms*—Multi-robot systems, coupled safety constraints, control barrier functions, safe reinforcement learning, equivariant graph neural networks, geometric deadlock, neighborhood proximal allocation.**

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I. Introduction

In cooperative inspection, on-demand wireless communication relaying, and dense formation reconfiguration, multi-robot systems must simultaneously satisfy two mutually conflicting classes of coupled scalar safety constraints: pairwise inter-robot collision avoidance, which pushes proximate robots apart, and global communication connectivity maintenance (represented by a scalar Laplacian algebraic connectivity lower bound), which couples the entire team together to prevent network fragmentation [1]–[3]. Control Barrier Functions (CBFs) [4], [5] provide a computationally efficient Quadratic Program (QP) safety filtering paradigm by translating forward invariance of a safe set into scalar affine inequality constraints on the control input. For multi-agent systems subject to globally and locally coupled scalar constraints, recent advances in continuous-time violation-free distributed gradient flows [6] and zero-sum Neighborhood Proximal Allocation (NPA) allow robots to compute strictly safe control inputs at each sampling instant using only a finite number of one-hop neighbor communication rounds.

However, instantaneous single-step feasibility of a safety filter does not imply long-horizon task completion. When multiple robots encounter one another in narrow passages or must exchange positions inside a tightly packed array, nominal attractive velocities pointing toward individual goals are frequently canceled by normal reaction forces from active collision and connectivity constraints, trapping the system permanently in non-target symmetric equilibria (i.e., geometric deadlocks) [7]–[10]. From a convex-optimization and physical perspective, traditional minimally invasive CBF filters lock up in dense encounters due to their myopic structure that enforces step-by-step monotone descent of the raw position error $W(\boldsymbol{p}) = \frac{1}{2}\sum_{i=1}^{N}\|\boldsymbol{p}_i - \boldsymbol{g}_i\|_2^2$. Projecting the negative-gradient nominal velocity $-k_g \nabla_{\boldsymbol{p}} W = -k_g(\boldsymbol{p} - \boldsymbol{g})$ onto a safe input set containing the zero-velocity vector necessarily enforces $\dot{W}(\boldsymbol{p}(t)) \leq 0$ [11]. In dense formations where interior

robots must swap positions while outer neighbors already reside at their respective goals ($\boldsymbol{p}_j = \boldsymbol{g}_j$), this monotonicity requirement induces a dual lock-up: first, the radial goal attraction of the swapping robots is collinearly canceled by the direct contact normal force; second, any temporary outward movement of a stationary outer neighbor to open a geometric passage momentarily increases $W(\boldsymbol{p})$, and under static equal quota splitting its local feasible set collapses to a singleton at zero velocity on the contact boundary [12].

To mitigate spurious equilibria and deadlocks in safety filters, two primary lines of research have been pursued: analytical circulation heuristics and learning-based safe control. On the analytical side, Reis et al. [7] and Tan and Dimarogonas [9] characterized undesired equilibria and saddle instabilities of CBF-QPs, while Gonçalves et al. [13] and Dai et al. [14] introduced tangential circulation terms and compatible Control Lyapunov–Barrier Functions (CLF-CBFs) to break collinear stagnation. Nevertheless, when interior swapping robots are surrounded by multiple layers of stationary target-holding neighbors, uniform constant-sign circulation triggers rotational conflicts across adjacent contact edges. More fundamentally, when a symmetric contact configuration constitutes a *strict constrained local minimum* rather than an unstable saddle point, no local negative-curvature perturbation or $\dot{W} \leq 0$ feedback can cross the surrounding energy barrier. On the learning side, primal-dual Lagrangian safe MARL (MAPPO-Lagrangian) [15], predictive and adaptive safety-filter reinforcement learning [16], [17], Neural Graph CBFs (GCBF+) [18], and Discrete Graph Proximal Policy Optimization (DGPPO) [19] employ Graph Neural Networks (GNNs) to learn multi-agent coordination and collision-avoidance policies. Despite their progress, existing safe Multi-Agent Reinforcement Learning (MARL) methods face three unresolved structural limitations:

1) **Lack of anytime zero-violation guarantees under global spectral connectivity constraints:** Neural barrier certificates and soft penalty rewards only handle pairwise relative-state collision constraints approximately and cannot guarantee deterministic, finite-round zero-violation invariance under global Laplacian algebraic connectivity constraints $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})) \geq \lambda_{\mathrm{req}}$ and sampled-data Zero-Order Hold (ZOH) execution.
2) **Disconnection between learned nominal policies and distributed quota allocation:** Existing safety layers typically rely on independent single-agent projections or static equal responsibility splitting ($B = 0$), which freezes stationary target-holding neighbors ($\boldsymbol{e}_j = \boldsymbol{0}$) and forces the nominal policy to explicitly coordinate yielding trajectories for all outer nodes, suffering from severe credit-assignment inefficiency.
3) **Terminal steady-state hovering induced by zero-order repulsive residuals:** Pairwise relative-direction biases output by message-passing GNNs are zero-order terms that do not vanish automatically as $\boldsymbol{e}_k \to \boldsymbol{0}$. On tightly packed target lattices with millimeter-scale clearance, persistent contact dual multipliers keep conflict gates open, causing steady-state limit-cycle hovering outside the goal tolerance ball.

To overcome these limitations, this paper proposes an *Envelope Equivariant Graph Policy with Neighborhood Proximal Allocation* (EGP-NPA) cooperative control architecture, which decomposes dense multi-robot deadlock resolution into three tightly coupled modules: a one-hop equivariant GNN residual nominal policy, a closed-neighborhood block dynamic quota-sharing shield, and a stateless one-hop closed-neighborhood completion envelope. The main contributions of this work are fourfold:

1) **Exact Rational Second-Order KKT Certificate of a Strict Non-Target Local Minimum (Theorem 2):** Using a rationally closed polynomial channel kernel and exact rational arithmetic ($\mathbb{Q}$), we construct an explicit primal–dual KKT witness on a seven-robot symmetric contact configuration where all robots are mobile. We prove that the Lagrangian Hessian is strictly positive definite over the entire critical cone ($\boldsymbol{v}^\top \boldsymbol{H}_L \boldsymbol{v} \geq \frac{19}{515}\|\boldsymbol{v}\|_2^2$) and establish a finite-radius quadratic energy barrier $W(\bar{\boldsymbol{p}}+\boldsymbol{\delta}) - W(\bar{\boldsymbol{p}}) \geq \frac{1}{100}\|\boldsymbol{\delta}\|_2^2 > 0$ for all collision-free perturbations $0 < \|\boldsymbol{\delta}\|_2 \leq \frac{1}{200}$ (which persists on an open ball $B_\rho(\boldsymbol{p}_0)$ in the strict interior of $\mathcal{S}$), alongside an explicit safe, connected reachability path ($\lambda_2(\boldsymbol{L}_{\mathrm{p}}) \geq 1792/625 > 5/2$). This certificate algebraically rules out saddle-escape or monotone-descent controllers and identifies two key physical mechanisms for deadlock resolution: lateral symmetry-breaking nominal drive from swapping robots and dynamic quota-driven yielding from stationary neighbors.
2) $\mathrm{SO}(2) \ltimes \mathbb{R}^2$ **and Permutation-Equivariant One-Hop GNN Residual Nominal Policy with Chiral Symmetry Breaking (Proposition 1):** We construct a parameter-shared 2-layer message-passing GNN over local target/edge orthogonal frames and compressed filter algebraic states (single-agent refit shadow prices $\boldsymbol{\mu}_{i,k-1}$, spectral quotas, and edge ledgers). We prove exact equivariance under planar rigid-body transformations $\mathrm{SO}(2) \ltimes \mathbb{R}^2$ and node permutations S_N, and prove both theoretically and empirically why preserving $\mathrm{SO}(2)$ chiral orientation is essential whereas enforcing full $\mathrm{O}(2)$ reflection equivariance forces tangential circulation $\omega_{i,k}^{\theta} \equiv 0$ on mirror-symmetric head-on configurations.
3) **Closed-Form Cooperative Unfreezing via Dynamic Quota Transfer and One-Hop Completion Envelope Theory (Propositions 2–4):** We prove anytime continuous-time hard safety invariance during noisy RL exploration and finite-round zero-shot cross-scale execution (Proposition 2). We derive the explicit closed-form solution for pairwise quota transfer $|t_{ij}^{\star}| = \frac{\eta d_{\mathrm{safe}}}{\eta + 4d_{\mathrm{safe}}^2}\delta_n > 0$ and stationary neighbor yielding velocity $-\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_j^{\star} = \frac{\eta}{2(\eta+4d_{\mathrm{safe}}^2)}\delta_n > 0$ driven by a moving robot's normal intrusion $\delta_n > 0$ (Proposition 3), proving why static equal splitting ($B = 0$) causes artificial freezing and why penalizing normal filter intervention suppresses $\delta_n > 0$. We further propose the one-hop closed-neighborhood completion envelope $\rho_{i,k}$, proving that it preserves cooperative residuals during transit while enforcing $\mathcal{O}(\|\boldsymbol{e}_k\|_2)$ first-order vanishing of zero-order pairwise bias at the goal

(Proposition 4).

4) **Small-Scale Short-Horizon Training, Zero-Shot Cross-Scale Generalization, and Second-Order Virtual-Physical Verification:** Trained via MAPPO only on small teams ($N \in \{6,\dots,10\}, B = 2$), EGP-NPA achieves $96/96$ (100%) arrival with 0 step cuts and 0 rational audit violations across 64 training-scale ($N = 8$) and 32 zero-shot cross-scale ($N \in \{12, 20, 25\}, B = 8$) dense lattice benchmarks, plus $48/48$ on an independent holdout suite. Controlled fixed-weight ablations and second-order `MuJoCo` omnidirectional chassis simulations coupled with an online division-free spectral estimator further validate the causal role of each module and establish the physical safety-margin versus channel-passability trade-off (Theorem 3).

Notation: $\mathbb{R}^n$, $\mathbb{Q}^n$, and $\mathbb{S}^n_+$ denote the n-dimensional real vector space, rational vector space, and $n \times n$ real symmetric positive semidefinite cone, respectively. $\boldsymbol{I}_n$ is the $n \times n$ identity matrix, $\mathbf{1}_N \in \mathbb{R}^N$ is the all-ones vector, and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^N$ is the i-th standard basis vector. $\|\cdot\|_2$ denotes the Euclidean vector norm or spectral matrix norm; $\mathrm{col}(\boldsymbol{x}_1,\dots,\boldsymbol{x}_m) = [\boldsymbol{x}_1^\top,\dots,\boldsymbol{x}_m^\top]^\top$; $\otimes$ is the Kronecker product; $(x)_+ = \max(x, 0)$; $\mathrm{clip}(x, a, b) = \min(\max(x, a), b)$; $\boldsymbol{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is the 90° counterclockwise rotation matrix satisfying $\boldsymbol{J}^\top = -\boldsymbol{J}$ and $\boldsymbol{J}^2 = -\boldsymbol{I}_2$; and $\mathbb{I}(\cdot)$ is the indicator function.

II. Problem Formulation and Preliminaries on Closed-Neighborhood NPA

A. Planar Multi-Robot Fixed-Goal Model and Rationally Closed Channel

Consider N mobile robots operating in the Euclidean plane $\mathbb{R}^2$ with vertex set $\mathcal{V} = \{1,\dots,N\}$. Each robot $i \in \mathcal{V}$ obeys velocity-bounded single-integrator kinematics:

$$\dot{\boldsymbol{p}}_i(t) = \boldsymbol{u}_i(t), \qquad \boldsymbol{u}_i(t) \in \mathcal{U}_i := \big\{\boldsymbol{v} \in \mathbb{R}^2 : \|\boldsymbol{v}\|_2 \le U\big\}, \tag{1}$$

where $\boldsymbol{p}_i \in \mathbb{R}^2$ and $\boldsymbol{u}_i \in \mathbb{R}^2$ are the planar position and velocity control input of robot i, and $U > 0$ is the maximum speed bound ($U = 0.50$ m/s throughout this paper). Given distinct static labeled goal positions $\boldsymbol{g}_i \in \mathbb{R}^2$ satisfying $\|\boldsymbol{g}_i - \boldsymbol{g}_j\|_2 \ge d_{\mathrm{safe}}$ ($\forall i \ne j$), define the stacked state $\boldsymbol{p} = \mathrm{col}(\boldsymbol{p}_1,\dots,\boldsymbol{p}_N) \in \mathbb{R}^{2N}$, joint input $\boldsymbol{u} = \mathrm{col}(\boldsymbol{u}_1,\dots,\boldsymbol{u}_N) \in \mathbb{R}^{2N}$, and goal configuration $\boldsymbol{g} = \mathrm{col}(\boldsymbol{g}_1,\dots,\boldsymbol{g}_N) \in \mathbb{R}^{2N}$. Let $\boldsymbol{e}_i = \boldsymbol{p}_i - \boldsymbol{g}_i$, $\boldsymbol{e} = \boldsymbol{p} - \boldsymbol{g}$, $\boldsymbol{p}_{ij} = \boldsymbol{p}_i - \boldsymbol{p}_j$, and $d_{ij} = \|\boldsymbol{p}_{ij}\|_2$. The raw navigation task distance cost is defined as

$$W(\boldsymbol{p}) = \frac{1}{2}\|\boldsymbol{e}\|_2^2 = \frac{1}{2}\sum_{i=1}^{N}\|\boldsymbol{p}_i - \boldsymbol{g}_i\|_2^2. \tag{2}$$

To rigorously separate intrinsic geometric deadlocks of the mathematical model from floating-point rounding artifacts during equilibrium certification and sampled-data ZOH verification, the physical communication graph employs a polynomial squared-cutoff channel kernel that is algebraically closed over the rational field $\mathbb{Q}$:

$$\omega(d) = \left(1 - \frac{d^2}{R_c^2}\right)_+^2, \qquad R_c = 10.0 \text{ m}. \tag{3}$$

Within the communication support $d < R_c$, the edge weight $w_{ij}(\boldsymbol{p}) = \omega(d_{ij})$ is a quadratic polynomial in the squared distance $d_{ij}^2 = \|\boldsymbol{p}_i - \boldsymbol{p}_j\|_2^2$, continuously differentiable at $d = R_c$ with locally Lipschitz gradient. With incidence vector $\boldsymbol{b}_{ij} = \boldsymbol{\varepsilon}_i - \boldsymbol{\varepsilon}_j \in \mathbb{R}^N$, the state-dependent weighted graph Laplacian is

$$\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}) = \sum_{1\le i<j\le N} \omega(d_{ij})\boldsymbol{b}_{ij}\boldsymbol{b}_{ij}^\top \in \mathbb{S}^N_+. \tag{4}$$

Whenever $\boldsymbol{p} \in \mathbb{Q}^{2N}$, the entries of $\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})$ and its directional derivatives are exact rational numbers, enabling zero-error verification of KKT optimality conditions and scalar safety margins via exact rational arithmetic.

B. Globally and Locally Coupled Scalar Safety Constraints and ZOH Segment Safety

Let $0 = \lambda_1(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})) \le \lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})) \le \cdots \le \lambda_N(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}))$ denote the ordered eigenvalues of $\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})$, where $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}))$ is the algebraic connectivity. The composite safe configuration set is $\mathcal{S} := \mathcal{X} \cap \mathcal{C}$, where $\mathcal{X} = [-B_{\mathrm{ws}}, B_{\mathrm{ws}}]^{2N}$ is the bounded workspace and

$$\begin{aligned}\mathcal{C} = \Big\{\boldsymbol{p} \in \mathbb{R}^{2N} : {} & h_0(\boldsymbol{p}) := \lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})) - \lambda_{\mathrm{req}} \ge 0, \\ & h_{ij}^{\mathrm{col}}(\boldsymbol{p}) := d_{ij}^2 - d_{\mathrm{safe}}^2 \ge 0 \ (\forall i < j)\Big\}\end{aligned} \tag{5}$$

is defined by one globally coupled scalar connectivity barrier function $h_0(\boldsymbol{p})$ and $P = N(N-1)/2$ pairwise coupled scalar collision-avoidance barrier functions $h_{ij}^{\mathrm{col}}(\boldsymbol{p})$ ($d_{\mathrm{safe}} > 0, \lambda_{\mathrm{req}} > 0$).

Focusing on deadlock resolution under coupled and mutually conflicting scalar constraints, we consider configurations with a simple Fiedler eigenvalue ($\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p})) < \lambda_3(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}))$). Let $\boldsymbol{\phi}(\boldsymbol{p}) \in \mathbb{R}^N$ ($\mathbf{1}_N^\top\boldsymbol{\phi}(\boldsymbol{p}) = 0, \|\boldsymbol{\phi}(\boldsymbol{p})\|_2 = 1$) denote the unit Fiedler eigenvector. By the Hellmann–Feynman theorem, the partial gradients of the scalar barrier functions $h_0(\boldsymbol{p})$ and $h_{ij}^{\mathrm{col}}(\boldsymbol{p})$ with respect to $\boldsymbol{p}_i$ are

$$\begin{aligned}\nabla_{\boldsymbol{p}_i} h_0(\boldsymbol{p}) &= \sum_{j\ne i}\big(\phi_i(\boldsymbol{p}) - \phi_j(\boldsymbol{p})\big)^2 \nabla_{\boldsymbol{p}_i} w_{ij}(\boldsymbol{p}), \\ \nabla_{\boldsymbol{p}_i} h_{ij}^{\mathrm{col}}(\boldsymbol{p}) &= -\nabla_{\boldsymbol{p}_j} h_{ij}^{\mathrm{col}}(\boldsymbol{p}) = 2\boldsymbol{p}_{ij}.\end{aligned} \tag{6}$$

Equation (6) highlights the geometric conflict between the two constraint classes: $\nabla_{\boldsymbol{p}_i} h_0(\boldsymbol{p})$ pulls connected robots inward to prevent network fragmentation, whereas $\nabla_{\boldsymbol{p}_i} h_{ij}^{\mathrm{col}}(\boldsymbol{p}) = 2\boldsymbol{p}_{ij}$ pushes proximate robots outward to prevent collisions. With barrier decay rates $\alpha_c, \alpha_s, \alpha_x > 0$, the continuous-time composite safe input set is defined by scalar affine CBF inequalities:

$$\begin{aligned}\mathcal{K}_0(\boldsymbol{p}) = \Big\{\boldsymbol{u} \in \mathbb{R}^{2N} : {} & 2\boldsymbol{p}_{ij}^\top(\boldsymbol{u}_i - \boldsymbol{u}_j) + \alpha_c h_{ij}^{\mathrm{col}}(\boldsymbol{p}) \ge 0, \\ & \sum_{i=1}^{N} \nabla_{\boldsymbol{p}_i} h_0(\boldsymbol{p})^\top \boldsymbol{u}_i + \alpha_s h_0(\boldsymbol{p}) \ge 0, \\ & \boldsymbol{u}_i \in \mathcal{U}_i, \ |p_{i,a} + \alpha_x^{-1} u_{i,a}| \le B_{\mathrm{ws}} \ (\forall i < j, a)\Big\}.\end{aligned} \tag{7}$$

For every $\boldsymbol{p} \in \mathcal{S}$, $\mathcal{K}_0(\boldsymbol{p})$ is nonempty, closed, and convex, and always contains the zero-velocity vector $\mathbf{0} \in \mathcal{K}_0(\boldsymbol{p})$. Given

a nominal velocity $\boldsymbol{u}^{\mathrm{nom}} \in \mathbb{R}^{2N}$, the centralized minimally invasive safety projection is

$$\Pi_{\mathcal{K}_0(\boldsymbol{p})}(\boldsymbol{u}^{\mathrm{nom}}) = \arg\min_{\boldsymbol{u}\in\mathcal{K}_0(\boldsymbol{p})} \frac{1}{2}\|\boldsymbol{u}-\boldsymbol{u}^{\mathrm{nom}}\|_2^2. \tag{8}$$

Under sampled-data ZOH execution at $t_k = k\Delta t$ ($\Delta t = 0.05$ s), the executed velocity $\boldsymbol{u}_k$ is held constant over $[t_k, t_k+h]$, generating the straight-line segment $\boldsymbol{p}(s) = \boldsymbol{p}_k + s\boldsymbol{u}_k$ ($s \in [0,h]$). Let $\bar{d}_{ij,k}(h) := \max\{d_{ij,k}, \|\boldsymbol{p}_{ij,k} + h(\boldsymbol{u}_{i,k}-\boldsymbol{u}_{j,k})\|_2\}$ denote the segment-wise maximum pairwise distance and $\underline{\boldsymbol{L}}_{\mathrm{p}}(\boldsymbol{p}_k, \boldsymbol{u}_k, h) := \sum_{i<j}\omega(\bar{d}_{ij,k}(h))\boldsymbol{b}_{ij}\boldsymbol{b}_{ij}^\top$ the corresponding constant lower-bound Laplacian; with the scalar Rayleigh-quotient second-order curvature constant $c_L = 8N/R_c^2$ of (3), we obtain the following continuous-time segment safety guarantee.

Theorem 1 (Continuous-Time Safety Over ZOH Segments)**.** *Suppose $\boldsymbol{p}_k \in \mathcal{S}$ and step size $h > 0$ satisfies $\alpha_c h \le 1, \alpha_s h \le 1, \alpha_x h \le 1$. If $\boldsymbol{u}_k$ satisfies the speed ball, workspace bounds, pairwise collision CBF inequalities in (7), and the scalar segment connectivity condition*

$$\begin{aligned} &\lambda_2\big(\underline{\boldsymbol{L}}_{\mathrm{p}}(\boldsymbol{p}_k, \boldsymbol{u}_k, h)\big) \ge \lambda_{\mathrm{req}} \quad \textit{or} \\ &\sum_{i=1}^{N} \nabla_{\boldsymbol{p}_i} h_0(\boldsymbol{p}_k)^\top \boldsymbol{u}_{i,k} + \alpha_s h_0(\boldsymbol{p}_k) \ge \frac{c_L h}{2}\|\boldsymbol{u}_k\|_2^2, \end{aligned} \tag{9}$$

then the continuous trajectory $\boldsymbol{p}(s) = \boldsymbol{p}_k + s\boldsymbol{u}_k$ satisfies $h_{ij}^{\mathrm{col}}(\boldsymbol{p}(s)) \ge 0$ ($\forall i<j$), $h_0(\boldsymbol{p}(s)) \ge 0$, and $\boldsymbol{p}(s) \in \mathcal{X}$ for all $s \in [0,h]$.

Proof. For each pair $i<j$, exact quadratic expansion of $h_{ij}^{\mathrm{col}}(\boldsymbol{p}_k + s\boldsymbol{u}_k)$ together with (7) and $\alpha_c s \le 1$ yields

$$\begin{aligned} h_{ij}^{\mathrm{col}}(\boldsymbol{p}_k + s\boldsymbol{u}_k) &= h_{ij}^{\mathrm{col}}(\boldsymbol{p}_k) + 2s\boldsymbol{p}_{ij,k}^\top(\boldsymbol{u}_{i,k}-\boldsymbol{u}_{j,k}) \\ &\quad + s^2\|\boldsymbol{u}_{i,k}-\boldsymbol{u}_{j,k}\|_2^2 \\ &\ge (1-\alpha_c s)h_{ij}^{\mathrm{col}}(\boldsymbol{p}_k) \ge 0. \end{aligned} \tag{10}$$

For the global scalar connectivity barrier $h_0(\boldsymbol{p}(s)) = \lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}(s))) - \lambda_{\mathrm{req}}$, convexity of $d_{ij}^2(s)$ on $[0,h]$ implies $\max_{s\in[0,h]} d_{ij}(s) = \bar{d}_{ij,k}(h)$, so for every unit vector $\boldsymbol{v} \in \mathbb{R}^N$ with $\mathbf{1}_N^\top \boldsymbol{v} = 0$, the scalar quadratic form $\psi_{\boldsymbol{v}}(s) := \boldsymbol{v}^\top \boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}_k + s\boldsymbol{u}_k)\boldsymbol{v}$ satisfies $\psi_{\boldsymbol{v}}(s) \ge \boldsymbol{v}^\top \underline{\boldsymbol{L}}_{\mathrm{p}}(\boldsymbol{p}_k, \boldsymbol{u}_k, h)\boldsymbol{v}$ across $s \in [0,h]$; thus the first condition in (9) yields $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}(s))) \ge \lambda_2(\underline{\boldsymbol{L}}_{\mathrm{p}}(\boldsymbol{p}_k, \boldsymbol{u}_k, h)) \ge \lambda_{\mathrm{req}}$ by the Courant–Fischer theorem. Similarly, applying second-order Taylor expansion with $\psi_{\boldsymbol{v}}''(\tau) \ge -c_L\|\boldsymbol{u}_k\|_2^2$ to $\psi_{\boldsymbol{v}}(s) - \lambda_{\mathrm{req}}$ and taking the infimum over unit zero-mean vectors $\boldsymbol{v}$ under the second condition in (9) gives

$$\begin{aligned} h_0(\boldsymbol{p}_k + s\boldsymbol{u}_k) &\ge (1-\alpha_s s)h_0(\boldsymbol{p}_k) \\ &\quad + \tfrac{1}{2}c_L s(h-s)\|\boldsymbol{u}_k\|_2^2 \ge 0. \end{aligned} \tag{11}$$

Workspace bounds hold for all $s \in [0,h]$ by convexity. □

C. Closed-Neighborhood Block NPA and Antisymmetric Edge-Ledger Settlement

At step k, the $M_k = 1+P_k$ coupled scalar safety constraints in (7) are written as affine inequality rows:

$$\sum_{i\in\mathcal{V}_a}\big(\boldsymbol{a}_{ia,k}^\top \boldsymbol{u}_i + b_{ia,k}\big) \le 0, \qquad a \in \{0,1,\dots,P_k\}, \tag{12}$$

where row $a=0$ ($\mathcal{V}_0 = \mathcal{V}$) is the globally coupled scalar connectivity row with unit Fiedler eigenvector $\boldsymbol{\phi}_k$ and discretization margin $\delta_\lambda > 0$ ($\lambda_{\mathrm{eff}} = \lambda_{\mathrm{req}} + \delta_\lambda$), i.e., $\boldsymbol{a}_{i0,k} = -\nabla_{\boldsymbol{p}_i} h_0(\boldsymbol{p}_k) = -\sum_{j\ne i}(\phi_{i,k}-\phi_{j,k})^2 \nabla_{\boldsymbol{p}_i} w_{ij}(\boldsymbol{p}_k)$ and level-clipped constant term $b_{i0,k} = -\frac{\alpha_s}{N}\big(\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}_k)) - \lambda_{\mathrm{eff}}\big)_+ \le 0$; rows $a \in \{1,\dots,P_k\}$ ($\mathcal{V}_a = \{i,j\}$) are pairwise scalar collision rows with $\boldsymbol{a}_{ia,k} = -2\boldsymbol{p}_{ij,k}, \boldsymbol{a}_{ja,k} = 2\boldsymbol{p}_{ij,k}$, and $b_{ia,k} = b_{ja,k} = -\frac{\alpha_c}{2}(d_{ij,k}^2 - d_{\mathrm{safe}}^2) \le 0$. Thus the zero-velocity feasible initialization condition $b_{ia,k} \le 0$ ($\forall a \in \{0,1,\dots,P_k\}$) holds identically throughout $\mathcal{S}$. In dense rearrangement tasks, the simultaneous enforcement of the global connectivity row and multiple tight pairwise contact rows at $d_{\mathrm{safe}} = 1.000$ m creates severe mutual conflict and geometric deadlocks.

Over the one-hop directed communication graph $\mathcal{G}_{\mathrm{c}}(\boldsymbol{p}_k) = (\mathcal{V}, \mathcal{E}_k)$ defined by sensing/communication radius $R_{\mathrm{sense}} = 2.50$ m (open neighborhood $\mathcal{N}_{i,k} = \{j : 0 < d_{ij,k} \le R_{\mathrm{sense}}\}$, closed neighborhood $\mathcal{N}_{i,k}^+ = \{i\} \cup \mathcal{N}_{i,k}$), each robot maintains zero-sum quota variables $s_{ia,k}$ reconstructed from antisymmetric edge ledgers $\varpi_{e,a,k}$ on undirected edges $e = \{i,j\}$:

$$s_{ia,k} = \sum_{e\ni i}\sigma_{i,e}\varpi_{e,a,k} \implies \sum_{i\in\mathcal{V}_a} s_{ia,k} = 0, \tag{13}$$

where $\sigma_{i,e} = +1, \sigma_{j,e} = -1$ for $i<j$. The closed-neighborhood NPA safety filter L executes two stages per control step:

1) **Stage 1 (Link-loss revocation, single-robot safety refit, and one-hop local settlement):** Edges exceeding R_{sense} have their ledgers zeroed by both endpoints. Each robot $i \in \mathcal{V}$ then solves the local refit QP under inherited quotas $\boldsymbol{s}_{i,k-1}$ (with nonnegative KKT multiplier vector $\boldsymbol{\mu}_{i,k-1}$):

$$\begin{aligned} \min_{\boldsymbol{u}_i\in\mathcal{U}_i\cap\mathcal{X}_i} \ & \frac{1}{2}\|\boldsymbol{u}_i - \boldsymbol{u}_{i,k}^{\mathrm{nom}}\|_2^2 \\ \text{s.t.} \ & \boldsymbol{a}_{ia,k}^\top \boldsymbol{u}_i + b_{ia,k} + s_{ia,k-1} \le 0, \ \forall a \ni i. \end{aligned} \tag{14}$$

If (14) is infeasible for robot i, one-hop local settlement clears all incident edge ledgers of node i ($s_{ia} = 0$, restoring feasibility of $\boldsymbol{u}_i = \mathbf{0}$ since $b_{ia,k} \le 0$) and notifies one-hop neighbors to re-solve (14), yielding a strictly feasible warm start $(\boldsymbol{u}_k^{(0)}, \boldsymbol{s}_k^{(0)})$.

2) **Stage 2 (B-round closed-neighborhood block proximal allocation):** In round $r \in \{1,\dots,B\}$, each closed-neighborhood induced subgraph $\mathcal{G}_i^+ = (\mathcal{N}_{i,k}^+, E_i)$ solves in parallel:

$$\begin{aligned} \min_{\tilde{\boldsymbol{u}}_q, w_q, t_e} \ & \frac{1}{2}\sum_{q\in\mathcal{N}_{i,k}^+}\|\tilde{\boldsymbol{u}}_q - \boldsymbol{u}_{q,k}^{\mathrm{nom}}\|_2^2 \\ & + \frac{1}{2\eta}\sum_{q\in\mathcal{N}_{i,k}^+} w_q^2 + \frac{1}{\eta}\sum_{e\in E_i} t_e^2 \\ \text{s.t.} \ & \boldsymbol{a}_{q0,k}^\top \tilde{\boldsymbol{u}}_q + b_{q0,k} + s_{q0,k}^{(r-1)} + w_q \le 0, \\ & \boldsymbol{a}_{qa,k}^\top \tilde{\boldsymbol{u}}_q + b_{qa,k} + s_{qa,k}^{(r-1)} + \sum_{e\in E_{i,a}}\sigma_{q,e}t_e \le 0, \\ & \sum_{q\in\mathcal{N}_{i,k}^+} w_q = 0, \quad \tilde{\boldsymbol{u}}_q \in \mathcal{U}_q \cap \mathcal{X}_q, \end{aligned} \tag{15}$$

where $\eta = 2.0$ is the proximal parameter, w_q is the block spectral quota increment, and t_e is the pairwise collision quota transfer on edge $e \in E_{i,a} = \{e \in E_i : a(e) = a\}$. Block solutions are committed via convex combination weights $\vartheta_i = (\max_{q\in\mathcal{N}^+_{i,k}} |\mathcal{N}^+_{q,k}|)^{-1}$:

$$\boldsymbol{u}_q^{(r)} = \Big(1 - \sum_{i:q\in\mathcal{N}^+_{i,k}} \vartheta_i\Big)\boldsymbol{u}_q^{(r-1)} + \sum_{i:q\in\mathcal{N}^+_{i,k}} \vartheta_i \tilde{\boldsymbol{u}}_q^{(i)}. \quad (16)$$

III. Obstruction of Myopic Error-Monotone Control and Rational KKT Certificate

A. Variational Inequality and Normal-Cone Lock-Up of Myopic Projections

In standard safety-critical navigation [4], [7], each robot uses the proportional goal-attraction nominal law $\boldsymbol{u}_i^{\mathrm{nom}} = -k_g \boldsymbol{e}_i = -k_g(\boldsymbol{p}_i - \boldsymbol{g}_i)$ ($k_g > 0$), yielding the filtered input $\boldsymbol{u}_G = \Pi_{\mathcal{K}_0(\boldsymbol{p})}(-k_g\boldsymbol{e})$. By the variational inequality of Euclidean projection onto the closed convex set $\mathcal{K}_0(\boldsymbol{p})$,

$$(\boldsymbol{u}_G + k_g\boldsymbol{e})^\top(\boldsymbol{v} - \boldsymbol{u}_G) \geq 0, \qquad \forall \boldsymbol{v} \in \mathcal{K}_0(\boldsymbol{p}). \quad (17)$$

Substituting the always-safe zero velocity $\boldsymbol{v} = \boldsymbol{0} \in \mathcal{K}_0(\boldsymbol{p})$ into (17) yields

$$\dot{W}(\boldsymbol{p}) = \boldsymbol{e}^\top \boldsymbol{u}_G \leq -\frac{1}{k_g}\|\boldsymbol{u}_G\|_2^2 \leq 0. \quad (18)$$

Equation (18) shows that whenever $\boldsymbol{0} \in \mathcal{K}_0(\boldsymbol{p})$, any proportional-attraction safety filter enforces monotone non-increasing $W(\boldsymbol{p}(t))$. However, at a non-target configuration ($\boldsymbol{e} \neq \boldsymbol{0}$), if $-k_g\boldsymbol{e}$ falls into the normal cone $N_{\mathcal{K}_0(\boldsymbol{p})}(\boldsymbol{0}) = \{\boldsymbol{w} \in \mathbb{R}^{2N} : \boldsymbol{w}^\top\boldsymbol{v} \leq 0,\ \forall \boldsymbol{v} \in \mathcal{K}_0(\boldsymbol{p})\}$ [20], no nonzero safe velocity can decrease W to first order, and the unique global optimum of the safety QP collapses to $\boldsymbol{u}_G = \boldsymbol{0}$.

B. Seven-Robot Strict Non-Target Local Minimum Theorem and Energy Barrier

Deadlocks in CBF filters are often treated as unstable saddle points escapable via small perturbations or negative curvature [7], [9]. To rigorously falsify the universality of this assumption when all robots are mobile, consider the seven-robot planar symmetric double-triangle configuration $\bar{\boldsymbol{p}} \in \mathbb{R}^{14}$ shown in Fig. 1(a) ($d_{\mathrm{safe}} = 1$ m, $U = 1/2$ m/s, columns correspond to robots $1, \ldots, 7$):

$$\bar{\boldsymbol{p}}^\top = \begin{bmatrix} -1 & -1/2 & -1/2 & 0 & 1/2 & 1/2 & 1 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 & 0 & \sqrt{3}/2 & -\sqrt{3}/2 & 0 \end{bmatrix}. \quad (19)$$

At $\bar{\boldsymbol{p}}$, all 7 robots are mobile and form 12 active contact edges at $d_{ij} = 1$ m: $\mathcal{E}_c = \{12, 13, 14, 24, 25, 34, 36, 45, 46, 47, 57, 67\}$. Outer robots $1, 2, 3, 5, 6$ start at their respective goals ($\boldsymbol{g}_i = \bar{\boldsymbol{p}}_i, \boldsymbol{e}_i = \boldsymbol{0}$), whereas center robot 4 at $(0, 0)$ and rightmost robot 7 at $(1, 0)$ must exchange positions ($\boldsymbol{g}_4 = \bar{\boldsymbol{p}}_7, \boldsymbol{g}_7 = \bar{\boldsymbol{p}}_4$).

Theorem 2 (Exact Rational Certificate of Strict Local Minimum and Energy Barrier)**.** *For the seven-robot position swap task (19) with $d_{\mathrm{safe}} = 1$ m, $\lambda_{\mathrm{req}} = 5/2$, $R_c = 10$ m:*

1) ***First-order stationarity and second-order strict positive definiteness on the critical cone:*** *At $\bar{\boldsymbol{p}}$, $W(\bar{\boldsymbol{p}}) = 1$, $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\bar{\boldsymbol{p}})) > 5/2$, and $\nabla W(\bar{\boldsymbol{p}}) = \frac{1}{2}\nabla h_{47}^{\mathrm{col}}(\bar{\boldsymbol{p}})$, so $\boldsymbol{u}_G = \boldsymbol{0}$ is the unique optimum of (8). Over the full critical cone $\mathcal{C} = \{\boldsymbol{v} \in \mathbb{R}^{14} : \bar{\boldsymbol{p}}_{ij}^\top(\boldsymbol{v}_i - \boldsymbol{v}_j) \geq 0\ (\forall\{i,j\} \in \mathcal{E}_c),\ v_{4x} = v_{7x}\}$, the Lagrangian Hessian $\boldsymbol{H}_L = \boldsymbol{I}_{14} - (\boldsymbol{b}_{47}\boldsymbol{b}_{47}^\top) \otimes \boldsymbol{I}_2$ satisfies the exact rational bounds:*

$$\min_{\boldsymbol{v}\in\mathcal{C},\ v_{4y}-v_{7y}=1} \|\boldsymbol{v}\|_2^2 = \frac{515}{496} > 1,$$
$$\boldsymbol{v}^\top \boldsymbol{H}_L \boldsymbol{v} \geq \frac{19}{515}\|\boldsymbol{v}\|_2^2 > 0 \quad (\forall \boldsymbol{v} \in \mathcal{C} \setminus \{\boldsymbol{0}\}). \quad (20)$$

2) ***Finite-radius spherical energy barrier and strictly interior open-ball trap:*** *For every collision-free perturbation $0 < \|\boldsymbol{\delta}\|_2 \leq \frac{1}{200}$ satisfying $d_{ij}(\bar{\boldsymbol{p}} + \boldsymbol{\delta}) \geq 1$ ($\forall i < j$),*

$$W(\bar{\boldsymbol{p}} + \boldsymbol{\delta}) - W(\bar{\boldsymbol{p}}) \geq \frac{1}{100}\|\boldsymbol{\delta}\|_2^2 > 0. \quad (21)$$

Moreover, for the strictly interior perturbed task $\boldsymbol{p}_0 = (1 + 10^{-8})\bar{\boldsymbol{p}}, \boldsymbol{g}_\tau = (1 + 10^{-6})\boldsymbol{g}$, every continuous safe trajectory starting in the open ball $B_\rho(\boldsymbol{p}_0) \subset \mathrm{int}(\mathcal{S})$ of radius $\rho = \frac{1}{4\times10^8}$ with $\dot{W}_\tau \leq 0$ is permanently trapped inside the sphere $\|\boldsymbol{p} - \bar{\boldsymbol{p}}\|_2 = \frac{1}{200}$ ($W_\tau(\bar{\boldsymbol{p}} + \boldsymbol{\delta}) - W_\tau(\bar{\boldsymbol{p}}) \geq \frac{19}{8\times10^7} > W_\tau(\boldsymbol{p}) - W_\tau(\bar{\boldsymbol{p}})$) and cannot reach the exterior goal $\boldsymbol{g}_\tau$.

3) ***Global collision-free and spectrally connected reachability:*** *There exists an explicit piecewise smooth path $\boldsymbol{p}(s)$ ($s \in [0, 3]$) connecting the initial and goal configurations such that $d_{ij}(s) \geq 1$ m and $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}(s))) \geq \frac{1792}{625} = 2.8672 > \frac{5}{2}$ for all $s \in [0, 3]$.*

Proof. (1) At $\bar{\boldsymbol{p}}$, $\boldsymbol{e}_4 = (-1, 0)^\top, \boldsymbol{e}_7 = (1, 0)^\top$, and $\boldsymbol{e}_i = \boldsymbol{0}$ otherwise, while $\nabla_{\boldsymbol{p}_4} h_{47}^{\mathrm{col}}(\bar{\boldsymbol{p}}) = (-2, 0)^\top$ and $\nabla_{\boldsymbol{p}_7} h_{47}^{\mathrm{col}}(\bar{\boldsymbol{p}}) = (2, 0)^\top$; thus multiplier $\lambda_{47} = 1/2$ (and 0 on all other edges) satisfies $\nabla W(\bar{\boldsymbol{p}}) - \frac{1}{2}\nabla h_{47}^{\mathrm{col}}(\bar{\boldsymbol{p}}) = \boldsymbol{0}$. For $\boldsymbol{v} \in \mathcal{C}$, $\boldsymbol{v}^\top\boldsymbol{H}_L\boldsymbol{v} = \|\boldsymbol{v}\|_2^2 - |v_{4y} - v_{7y}|^2$. Scaling coordinates as $\boldsymbol{v}_i = (z_{ix}, \sqrt{3}z_{iy})^\top$ stacked into $\boldsymbol{z} \in \mathbb{R}^{14}$ with $Q_c = \mathrm{diag}(1, 3, \ldots, 1, 3)$ gives $\|\boldsymbol{v}\|_2^2 = \boldsymbol{z}^\top Q_c \boldsymbol{z}$. Define rational matrices $(G_c\boldsymbol{z})_{ij} = \bar{p}_{ij,x}(z_{ix} - z_{jx}) + \sqrt{3}\bar{p}_{ij,y}(z_{iy} - z_{jy})$ and $E_c\boldsymbol{z} = [z_{4x} - z_{7x},\ z_{4y} - z_{7y}]^\top$. For the convex QP $\min \boldsymbol{z}^\top Q_c \boldsymbol{z}$ s.t. $G_c\boldsymbol{z} \geq \boldsymbol{0}$, $E_c\boldsymbol{z} = (0, 1)^\top$, the exact rational primal witness $Z_* \in \mathbb{Q}^{7\times2}$ and dual multipliers $(\boldsymbol{\mu}, \boldsymbol{\nu})$ are:

$$Z_*^\top = \frac{1}{496}\begin{bmatrix} -15 & -132 & -105 & 90 & 177 & -105 & 90 \\ 15 & 132 & -15 & 206 & 177 & -225 & -290 \end{bmatrix},$$
$$\mu_{13} = \tfrac{15}{248},\ \mu_{24} = \tfrac{33}{62},\ \mu_{36} = \tfrac{15}{62},\ \mu_{45} = \tfrac{177}{248},\ \mu_{67} = \tfrac{225}{248},$$
$$\boldsymbol{\nu} = \left[\tfrac{135}{496},\ \tfrac{1545}{496}\right]^\top, \quad (22)$$

with all other $\mu_{ij} = 0$. Exact rational arithmetic verifies primal feasibility $G_c\boldsymbol{z}_* \geq \boldsymbol{0}, E_c\boldsymbol{z}_* = (0, 1)^\top$, complementary slackness $\boldsymbol{\mu}^\top G_c\boldsymbol{z}_* = 0$, and stationarity $Q_c\boldsymbol{z}_* = G_c^\top\boldsymbol{\mu} + E_c^\top\boldsymbol{\nu}$, with optimal objective $\boldsymbol{z}_*^\top Q_c \boldsymbol{z}_* = \frac{1545}{496}$. Dividing by the vertical scaling factor 3 yields $\min_{\boldsymbol{v}\in\mathcal{C}, v_{4y}-v_{7y}=1}\|\boldsymbol{v}\|_2^2 = \frac{515}{496} > 1$ and $\boldsymbol{v}^\top\boldsymbol{H}_L\boldsymbol{v} \geq (1 - \frac{496}{515})\|\boldsymbol{v}\|_2^2 = \frac{19}{515}\|\boldsymbol{v}\|_2^2$.

(2) Let $0 < R := \|\boldsymbol{\delta}\|_2 \leq \frac{1}{200}$ satisfy $\|\bar{\boldsymbol{p}}_{ij} + \boldsymbol{\delta}_i - \boldsymbol{\delta}_j\|_2^2 \geq 1$ ($\forall\{i,j\} \in \mathcal{E}_c$). With $z = \delta_{4x} - \delta_{7x}$ and $t = \delta_{4y} - \delta_{7y}$, we have $W(\bar{\boldsymbol{p}} + \boldsymbol{\delta}) - W(\bar{\boldsymbol{p}}) = \frac{1}{2}R^2 - z$. If $z \leq 0$, (21) holds trivially; suppose $z > 0$. Each contact edge gives $\bar{\boldsymbol{p}}_{ij}^\top(\boldsymbol{\delta}_i - \boldsymbol{\delta}_j) \geq -\frac{1}{2}\|\boldsymbol{\delta}_i - \boldsymbol{\delta}_j\|_2^2 \geq -R^2$. Taking the inner product

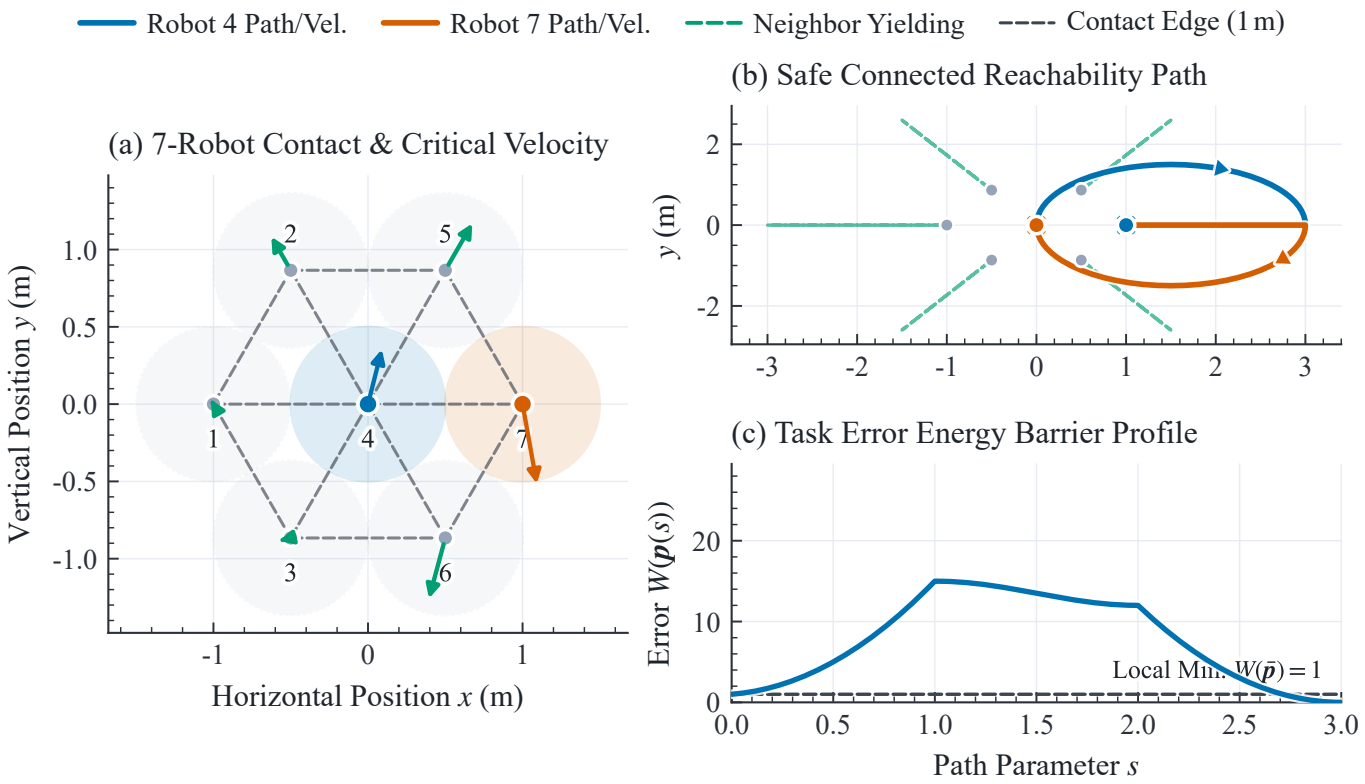


Fig. 1. Seven-robot strict non-target local minimum configuration and analytical safe connected reachability witness. (a) Double-triangle contact configuration (19), 12 initial contact edges, and optimal critical-cone velocity witness vectors (22); (b) analytical 3-stage reachability path (uniform expansion by 3, opposite semicircular swap of robots 4 and 7, and uniform contraction); (c) task error profile $W(\boldsymbol{p}(s))$ along the witness path, illustrating the mandatory energy barrier above $W(\bar{\boldsymbol{p}}) = 1$.

of $Q_c \boldsymbol{z}_* = G_c^\top \boldsymbol{\mu} + E_c^\top \boldsymbol{\nu}$ with the scaled perturbation $\tilde{\boldsymbol{\delta}}$, using $\sum \mu_{ij} = \frac{609}{248} =: S$, $c = \frac{1545}{496}$, and the Cauchy–Schwarz inequality gives $\frac{c}{\sqrt{3}}|t| \le \sqrt{c}R + SR^2 \implies t^2 \le \frac{496}{515}R^2(1 + \frac{3}{2}R)^2$. Moreover, $(-1+z)^2 + t^2 \ge 1 \implies 2z \le z^2 + t^2$ and $z \le 2R^2$, whence

$$\begin{aligned} W(\bar{\boldsymbol{p}} + \boldsymbol{\delta}) - W(\bar{\boldsymbol{p}}) &= \frac{1}{2}R^2 - z \\ &\ge \frac{R^2}{2}\left[1 - \frac{496}{515}\left(1 + \frac{3}{2}R\right)^2 - 4R^2\right] \\ &\ge \frac{1}{100}R^2, \qquad \forall R \in \left(0, \frac{1}{200}\right]. \end{aligned} \tag{23}$$

The strictly interior trap on $B_\rho(\boldsymbol{p}_0)$ follows from $W_\tau(\bar{\boldsymbol{p}}+\boldsymbol{\delta}) - W_\tau(\bar{\boldsymbol{p}}) \ge \frac{r^2}{100} - \frac{5}{2}\tau r = \frac{19}{8\times 10^7}$ on $\|\boldsymbol{\delta}\|_2 = r = \frac{1}{200}$.

(3) Along the 3-stage path in Fig. 1(b) (uniform expansion to $3\bar{\boldsymbol{p}}$, opposite semicircular swap of robots 4 and 7 around $(3/2, 0)$ with radius $3/2$, and uniform contraction to $\boldsymbol{g}$), all pairwise distances satisfy $d_{ij}(s) \ge 1$ and $\max_{i<j} d_{ij}(s) \le 6$, so $w_{ij}(s) \ge (1 - 36/100)^2 = 256/625$ and $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}(s))) \ge 7 \times \frac{256}{625} = \frac{1792}{625} > \frac{5}{2}$. □

Remark 1 (Cooperative Control Constructions Motivated by the Witness)**.** *The exact rational witness (22) motivates two local control constructions used here. These are not global necessary and sufficient conditions for arbitrary rearrangement tasks:*

1) ***Lateral symmetry-breaking nominal drive from swapping robots:*** *In (22), robots 4 and 7 must generate opposite transverse velocity components $z_{4y} = +\frac{206}{496} > 0$ and $z_{7y} = -\frac{290}{496} < 0$ perpendicular to $\boldsymbol{e}_4, \boldsymbol{e}_7$. Purely radial attraction keeps the system inside the collinear deadlock subspace $v_{4y} - v_{7y} = 0$.*
2) ***Cooperative yielding of outer stationary neighbors via dynamic quota transfer:*** *As robots 4 and 7 separate transversely, the five outer target-holding neighbors ($\boldsymbol{e}_j = \boldsymbol{0}$) must simultaneously execute nonzero outward yielding velocities (e.g., $(Z_*)_{2,:} = \frac{1}{496}[-132, 132]$, $(Z_*)_{5,:} = \frac{1}{496}[177, 177]$), transmitted mechanically through strictly positive contact shadow prices $\mu_{24}, \mu_{45}, \mu_{67} > 0$. Under static equal quota splitting ($s_{ia} \equiv 0$), stationary neighbors remain frozen at $\boldsymbol{u}_j = \boldsymbol{0}$ and truncate the transverse separation of robots 4 and 7 to zero.*

IV. Envelope Equivariant Graph Policy with Neighborhood Proximal Allocation (EGP-NPA)

A. Chiral Rotation- and Permutation-Equivariant GNN Residual Policy

To break the collinear deadlock identified in Remark 1 without hand-crafting global topological potentials, we construct a distributed residual nominal policy over the one-hop directed communication graph $\mathcal{G}_{\mathrm{c}}(\boldsymbol{p}_k) = (\mathcal{V}, \mathcal{E}_k)$ at $t_k = k\Delta t$ ($R_{\mathrm{sense}} = 2.50$ m).

Feeding raw Cartesian coordinates $\boldsymbol{p}_{i,k}, \boldsymbol{g}_i$ into a neural network violates Euclidean rigid-body invariance, causing identical relative formations rotated or translated in the workspace to produce inconsistent actions [21]. For each robot $i \in \mathcal{V}$, we construct a local target orthonormal frame using $\hat{\boldsymbol{e}}_{i,k} = \boldsymbol{e}_{i,k}/\max(\|\boldsymbol{e}_{i,k}\|_2, 10^{-9})$ and $\boldsymbol{J}\hat{\boldsymbol{e}}_{i,k}$, together with a smooth goal-vanishing factor $\sigma_{i,k} = \min(1, \|\boldsymbol{e}_{i,k}\|_2/r_0)$ ($r_0 = 0.10$ m) to remove directional singularity as $\boldsymbol{e}_{i,k} \to \boldsymbol{0}$. For each directed edge $(i,j) \in \mathcal{E}_k$, we construct a local edge orthonormal frame using $\hat{\boldsymbol{r}}_{ij,k} = (\boldsymbol{p}_{j,k} - \boldsymbol{p}_{i,k})/d_{ij,k}$ and $\boldsymbol{J}\hat{\boldsymbol{r}}_{ij,k}$.

Let $\boldsymbol{u}_{i,k-1}$ be the previous executed velocity, $\boldsymbol{\delta}_{i,k-1} = \boldsymbol{u}_{i,k-1} - \Pi_{\mathcal{U}_i}(\boldsymbol{u}^{\mathrm{nom}}_{i,k-1})$ the filter velocity modification, $\boldsymbol{\mu}_{i,k-1}$ the KKT dual multiplier vector (shadow prices) from the single-robot refit QP (14), $s_{i0,k-1}$ the spectral quota, and $\boldsymbol{\varpi}_{e,k-1} = [\varpi_{ij,0,k-1}, \varpi_{ij,1,k-1}]^\top$ the binary edge ledger on $e = \{i,j\}$. Define the one-hop closed-neighborhood conflict memory $m_{i,k} = \max\big(\gamma_m m_{i,k-1}, \max_{j\in\mathcal{N}^+_{i,k}} c_{j,k-1}\big)$ and gate $\gamma^{\mathrm{gate}}_{i,k} = \tanh(\beta_m m_{i,k}) \in [0,1)$ ($\gamma_m = 0.97, \beta_m = 4.0$; $c_{j,k-1} = \|\boldsymbol{\mu}_{j,k-1}\|_2$ gives the shadow-price gate $\gamma^{\mathrm{gate},\mu}_{i,k}$, while $c_{j,k-1} = \|\boldsymbol{\delta}_{j,k-1}\|_2$ gives the intervention gate $\gamma^{\mathrm{gate,int}}_{i,k}$). The rotation-invariant node feature $\boldsymbol{x}_{i,k} = \mathrm{col}(\boldsymbol{x}^{\mathrm{geom}}_{i,k}, \boldsymbol{x}^{\mathrm{alg}}_{i,k}) \in \mathbb{R}^{17}$ and directed edge feature $\boldsymbol{e}^{\mathrm{feat}}_{ij,k} = \mathrm{col}(\boldsymbol{e}^{\mathrm{geom}}_{ij,k}, \boldsymbol{e}^{\mathrm{alg}}_{ij,k}) \in \mathbb{R}^{20}$ are constructed as:

$$\begin{aligned}\boldsymbol{x}^{\mathrm{geom}}_{i,k} = \Big[&\min(\|\boldsymbol{e}_{i,k}\|_2, 4),\ \mathbb{I}(\|\boldsymbol{e}_{i,k}\|_2 < \varepsilon_g),\ \gamma^{\mathrm{gate}}_{i,k},\\ &\tfrac{\|\boldsymbol{u}_{i,k-1}\|_2}{U},\ \tfrac{\boldsymbol{u}^\top_{i,k-1}\hat{\boldsymbol{e}}_{i,k}}{U}\sigma_{i,k},\ \tfrac{\boldsymbol{u}^\top_{i,k-1}\boldsymbol{J}\hat{\boldsymbol{e}}_{i,k}}{U}\sigma_{i,k},\\ &\tfrac{\|\boldsymbol{\delta}_{i,k-1}\|_2}{U},\ \tfrac{\boldsymbol{\delta}^\top_{i,k-1}\hat{\boldsymbol{e}}_{i,k}}{U}\sigma_{i,k},\ \tfrac{\boldsymbol{\delta}^\top_{i,k-1}\boldsymbol{J}\hat{\boldsymbol{e}}_{i,k}}{U}\sigma_{i,k},\\ &\tfrac{\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}_k))-\lambda_{\mathrm{req}}}{\lambda_{\mathrm{req}}},\ \tfrac{|\mathcal{N}_{i,k}|}{8},\ \min_{j\in\mathcal{N}_{i,k}}(d_{ij,k} - d_{\mathrm{safe}})\Big]^\top,\end{aligned} \tag{24}$$

$$\begin{aligned}\boldsymbol{e}^{\mathrm{geom}}_{ij,k} = \Big[&\tfrac{d_{ij,k}}{R_{\mathrm{sense}}},\ \mathrm{clip}(d_{ij,k} - d_{\mathrm{safe}}, 0, 1.5),\ \hat{\boldsymbol{r}}^\top_{ij,k}\hat{\boldsymbol{e}}_{i,k}\sigma_{i,k},\\ &\hat{\boldsymbol{r}}^\top_{ij,k}\boldsymbol{J}\hat{\boldsymbol{e}}_{i,k}\sigma_{i,k},\ \hat{\boldsymbol{r}}^\top_{ij,k}\hat{\boldsymbol{e}}_{j,k}\sigma_{j,k},\ \hat{\boldsymbol{r}}^\top_{ij,k}\boldsymbol{J}\hat{\boldsymbol{e}}_{j,k}\sigma_{j,k},\\ &\min(\|\boldsymbol{e}_{j,k}\|_2, 4),\ \hat{\boldsymbol{e}}^\top_{i,k}\hat{\boldsymbol{e}}_{j,k}\sigma_{i,k}\sigma_{j,k},\\ &(\boldsymbol{J}\hat{\boldsymbol{e}}_{i,k})^\top\hat{\boldsymbol{e}}_{j,k}\sigma_{i,k}\sigma_{j,k},\ \tfrac{(\boldsymbol{u}_{j,k-1}-\boldsymbol{u}_{i,k-1})^\top\hat{\boldsymbol{r}}_{ij,k}}{U},\\ &\tfrac{(\boldsymbol{u}_{j,k-1}-\boldsymbol{u}_{i,k-1})^\top\boldsymbol{J}\hat{\boldsymbol{r}}_{ij,k}}{U},\ \gamma^{\mathrm{gate}}_{j,k},\ \tfrac{\|\boldsymbol{\delta}_{j,k-1}\|_2}{U}\Big]^\top,\end{aligned} \tag{25}$$

$$\begin{aligned}\boldsymbol{x}^{\mathrm{alg}}_{i,k} = \Big[&\ln(1+|\mu_{i0,k-1}|),\ \ln\big(1+\textstyle\sum_{j\neq i}|\mu_{ij,k-1}|\big),\\ &\mathrm{clip}\big(\tfrac{s_{i0,k-1}}{\bar{\nu}_{i0,k}}, -3, 3\big),\ \ln(1+\|\boldsymbol{\mu}_{i,k-1}\|_2),\\ &\mathrm{clip}\big(\tfrac{U\|\boldsymbol{a}_{i0,k}\|_2 - b_{i0,k} - s_{i0,k-1}}{\bar{\nu}_{i0,k}}, -3, 3\big)\Big]^\top,\\ \boldsymbol{e}^{\mathrm{alg}}_{ij,k} = \Big[&\ln(1+|\mu_{ij,k-1}|),\ \ln(1+|\mu_{ji,k-1}|),\\ &\tanh(2\varpi_{ij,0,k-1}),\ \tanh(2\varpi_{ij,1,k-1}),\\ &|\tilde{\phi}_{i,k} - \tilde{\phi}_{j,k}|,\ \|\widetilde{\boldsymbol{\Phi}}_{i,k} - \widetilde{\boldsymbol{\Phi}}_{j,k}\|_2,\ \widetilde{\boldsymbol{\Phi}}^\top_{i,k}\widetilde{\boldsymbol{\Phi}}_{j,k}\Big]^\top,\end{aligned} \tag{26}$$

where $\bar{\nu}_{i0,k} = U\|\boldsymbol{a}_{i0,k}\|_2 + |b_{i0,k}| + 10^{-3}$ normalizes the global scalar constraint row, and $\tilde{\phi}_{i,k} = \sqrt{N}\phi_{i,k}$, $\widetilde{\boldsymbol{\Phi}}_{i,k} = \sqrt{N}\boldsymbol{\Phi}_{i,k} \in \mathbb{R}^2$ are unit-RMS normalized scalar Laplacian spectral coordinates ($\frac{1}{N}\sum_{i=1}^N \tilde{\phi}^2_{i,k} = 1$) that encode scale-free relative topological positions via local differences.

A parameter-shared $L_{\mathrm{gnn}} = 2$-layer message-passing GNN encodes $\boldsymbol{h}^{(0)}_i = \phi_{\mathrm{enc}}(\boldsymbol{x}_{i,k}) \in \mathbb{R}^{d_h}$ ($d_h = 64$) and performs one-hop message passing with mean–max dual aggregation across layers $\ell \in \{1, \dots, L_{\mathrm{gnn}}\}$:

$$\begin{aligned}\boldsymbol{m}^{(\ell)}_{ij} &= \phi^{(\ell)}_{\mathrm{msg}}\big(\mathrm{col}(\boldsymbol{h}^{(\ell-1)}_i, \boldsymbol{h}^{(\ell-1)}_j, \boldsymbol{e}^{\mathrm{feat}}_{ij,k})\big), \quad \forall (i,j) \in \mathcal{E}_k,\\ \bar{\boldsymbol{m}}^{(\ell)}_i &= \frac{1}{\max(|\mathcal{N}_{i,k}|, 1)}\sum_{j\in\mathcal{N}_{i,k}} \boldsymbol{m}^{(\ell)}_{ij}, \quad \hat{\boldsymbol{m}}^{(\ell)}_i = \max_{j\in\mathcal{N}_{i,k}} \boldsymbol{m}^{(\ell)}_{ij},\\ \boldsymbol{h}^{(\ell)}_i &= \mathrm{LN}^{(\ell)}\Big(\boldsymbol{h}^{(\ell-1)}_i + \phi^{(\ell)}_{\mathrm{upd}}\big(\mathrm{col}(\boldsymbol{h}^{(\ell-1)}_i, \bar{\boldsymbol{m}}^{(\ell)}_i, \hat{\boldsymbol{m}}^{(\ell)}_i)\big)\Big),\end{aligned} \tag{27}$$

where $\phi_{\mathrm{enc}}, \phi^{(\ell)}_{\mathrm{msg}}, \phi^{(\ell)}_{\mathrm{upd}}$ are Multi-Layer Perceptrons (MLPs) with SiLU activations and $\mathrm{LN}^{(\ell)}$ denotes Layer Normalization. Output heads ϕ_{node} and ϕ_{edge} produce bounded scalar action coefficients:

$$\begin{aligned}(\omega^\theta_{i,k}, r^\theta_{i,k}) &= \mathrm{clip}\big(\phi_{\mathrm{node}}(\boldsymbol{h}^{(L_{\mathrm{gnn}})}_i), -1, 1\big),\\ \psi^\theta_{ij,k} &= \mathrm{clip}\big(\phi_{\mathrm{edge}}(\mathrm{col}(\boldsymbol{h}^{(L_{\mathrm{gnn}})}_i, \boldsymbol{h}^{(L_{\mathrm{gnn}})}_j, \boldsymbol{e}^{\mathrm{feat}}_{ij,k})), -1, 1\big),\end{aligned} \tag{28}$$

which synthesize the planar nominal velocity of robot i:

$$\begin{aligned}\boldsymbol{u}^{\mathrm{nom}}_{i,k}(\boldsymbol{p}_k;\theta) &= -\boldsymbol{e}_{i,k} + \gamma^{\mathrm{eff}}_{i,k}\boldsymbol{r}^\theta_{i,k}(\boldsymbol{p}_k),\\ \boldsymbol{r}^\theta_{i,k}(\boldsymbol{p}_k) &= A_{\mathrm{rot}}\omega^\theta_{i,k}\boldsymbol{J}\boldsymbol{e}_{i,k} + A_{\mathrm{rad}}r^\theta_{i,k}\boldsymbol{e}_{i,k}\\ &\quad + \frac{A_\psi U}{\max(|\mathcal{N}_{i,k}|, 1)}\sum_{j\in\mathcal{N}_{i,k}} \psi^\theta_{ij,k}\frac{\boldsymbol{p}_{i,k} - \boldsymbol{p}_{j,k}}{d_{ij,k}},\end{aligned} \tag{29}$$

where $\gamma^{\mathrm{eff}}_{i,k} \in [0,1)$ is the composite envelope gate (Section IV-C) and $A_{\mathrm{rot}} = 2.0\ \mathrm{s}^{-1}, A_{\mathrm{rad}} = 1.5\ \mathrm{s}^{-1}, A_\psi = 2.0$. Here $-\boldsymbol{e}_{i,k}$ provides baseline goal attraction; $A_{\mathrm{rot}}\omega^\theta_{i,k}\boldsymbol{J}\boldsymbol{e}_{i,k}$ generates tangential circulation perpendicular to the line-of-sight to break head-on collinear symmetry; $A_{\mathrm{rad}}r^\theta_{i,k}\boldsymbol{e}_{i,k}$ modulates radial approach speed; and $\sum_{j\in\mathcal{N}_{i,k}}\psi^\theta_{ij,k}(\boldsymbol{p}_{i,k} - \boldsymbol{p}_{j,k})/d_{ij,k}$ provides pairwise repulsive bias along neighbor links.

Proposition 1 ($\mathrm{SO}(2) \ltimes \mathbb{R}^2$ and Permutation Equivariance and Chiral Symmetry Breaking)**.** *For any network parameters θ, any rotation matrix $\boldsymbol{R} \in \mathrm{SO}(2)$, translation $\boldsymbol{c} \in \mathbb{R}^2$, and $N \times N$ permutation matrix $\boldsymbol{P}_\pi$, let $(\tilde{\boldsymbol{p}}_i, \tilde{\boldsymbol{g}}_i, \tilde{\boldsymbol{u}}_{i,k-1}) = (\boldsymbol{R}\boldsymbol{p}_i + \boldsymbol{c}, \boldsymbol{R}\boldsymbol{g}_i + \boldsymbol{c}, \boldsymbol{R}\boldsymbol{u}_{i,k-1})$. Then the nominal control field (29) satisfies:*

$$\begin{aligned}\boldsymbol{u}^{\mathrm{nom}}_{i,k}(\tilde{\boldsymbol{p}}_k, \tilde{\boldsymbol{g}}; \theta) &= \boldsymbol{R}\boldsymbol{u}^{\mathrm{nom}}_{i,k}(\boldsymbol{p}_k, \boldsymbol{g}; \theta),\\ \boldsymbol{u}^{\mathrm{nom}}_k\big((\boldsymbol{P}_\pi \otimes \boldsymbol{I}_2)\boldsymbol{p}_k, (\boldsymbol{P}_\pi \otimes \boldsymbol{I}_2)\boldsymbol{g}\big) &= (\boldsymbol{P}_\pi \otimes \boldsymbol{I}_2)\boldsymbol{u}^{\mathrm{nom}}_k(\boldsymbol{p}_k, \boldsymbol{g}).\end{aligned} \tag{30}$$

Furthermore, if (29) is constrained to be equivariant under the full orthogonal group $\mathrm{O}(2)$ (including planar reflections $\boldsymbol{M} \in \mathrm{O}(2) \setminus \mathrm{SO}(2)$ with $\det \boldsymbol{M} = -1$), then on any encounter configuration that is mirror-symmetric across $\hat{\boldsymbol{e}}_{i,k}$, the tangential circulation coefficient necessarily collapses to $\omega^\theta_{i,k} \equiv 0$.

Proof. Under $\mathrm{SO}(2) \ltimes \mathbb{R}^2$, $\tilde{\boldsymbol{e}}_{i,k} = \boldsymbol{R}\boldsymbol{e}_{i,k}$ and $\tilde{\boldsymbol{p}}_{ij,k} = \boldsymbol{R}\boldsymbol{p}_{ij,k}$. Since $\boldsymbol{R} \in \mathrm{SO}(2)$ preserves Euclidean norms and inner products and commutes with $\boldsymbol{J}$ ($\boldsymbol{R}\boldsymbol{J} = \boldsymbol{J}\boldsymbol{R}$), all scalar features in (24)–(26) and action coefficients $(\omega^\theta_{i,k}, r^\theta_{i,k}, \psi^\theta_{ij,k})$ are strictly

invariant. Substituting into (29) and using $\boldsymbol{R}\boldsymbol{J}\boldsymbol{e}_{i,k} = \boldsymbol{J}\boldsymbol{R}\boldsymbol{e}_{i,k}$ yields the first identity; permutation equivariance follows from parameter sharing and symmetric mean/max aggregation in (27). Conversely, any reflection $\boldsymbol{M} \in \mathrm{O}(2) \setminus \mathrm{SO}(2)$ anticommutes with $\boldsymbol{J}$ ($\boldsymbol{M}\boldsymbol{J} = -\boldsymbol{J}\boldsymbol{M}$). On a state mirror-symmetric under $\boldsymbol{M}$ ($\boldsymbol{M}\boldsymbol{e}_{i,k} = \boldsymbol{e}_{i,k}$), $\mathrm{O}(2)$ equivariance requires $\boldsymbol{M}\boldsymbol{u}_{i,k}^{\mathrm{nom}} = \boldsymbol{u}_{i,k}^{\mathrm{nom}}$, whose projection along $\boldsymbol{J}\boldsymbol{e}_{i,k}$ forces $-\omega_{i,k}^{\theta} = \omega_{i,k}^{\theta} \implies \omega_{i,k}^{\theta} \equiv 0$. □

B. Closed-Neighborhood Dynamic Quota Shield and Cooperative Yielding of Stationary Neighbors

The nominal velocity $\boldsymbol{u}_{i,k}^{\mathrm{nom}}$ is filtered by the B-round closed-neighborhood block NPA shield L (Section II-C) and verified by the exact rational ZOH segment certificate (Theorem 1).

Proposition 2 (Anytime Hard Safety Invariance Under Exploration and Cross-Scale Deployment)**.** *Suppose $\boldsymbol{p}_0 \in \mathcal{S}$ and $b_{ia,k} \le 0$. Then for arbitrary policy parameters θ (including noisy policies during training exploration and zero-shot deployment to any team size N) and any finite number of allocation rounds $B \ge 0$, the closed-loop continuous trajectory $\boldsymbol{p}(t)$ satisfies $d_{ij}(t) \ge d_{\mathrm{safe}}$ ($\forall i < j$), $\lambda_2(\boldsymbol{L}_{\mathrm{p}}(\boldsymbol{p}(t))) \ge \lambda_{\mathrm{req}}$, $\boldsymbol{u}_i(t) \in \mathcal{U}_i$, and $\boldsymbol{p}(t) \in \mathcal{X}$ for all $t \ge 0$.*

Proof. By the antisymmetric edge-ledger construction (13), Stage 1 revocation and one-hop local settlement guarantee $\sum_{i \in \mathcal{V}_a} s_{ia,k}^{(0)} = 0$ and single-robot feasibility (14). Each block QP (15) in Stage 2 preserves zero-sum quotas via $\sum_{q \in \mathcal{N}_{i,k}^{+}} w_q = 0$ and $\sigma_{q,e} t_e + \sigma_{m,e} t_e = 0$, and the convex combination (16) maintains feasibility of $(\boldsymbol{u}_k^{(B)}, \boldsymbol{s}_k^{(B)})$ for all coupled constraints (12). Combined with adaptive ZOH step-halving under Theorem 1, $\boldsymbol{p}(t) \in \mathcal{S}$ holds for all $t \ge 0$ by induction. □

We next prove algebraically why dynamic quota transfer t_e in (15) is the exact mechanism that unfreezes outer stationary neighbors, whereas static equal allocation ($B = 0$) induces artificial freezing.

Proposition 3 (Cooperative Unfreezing of Stationary Blocking Neighbors via Dynamic Quota Transfer)**.** *Consider a contact pair on the safety boundary $d_{ij} = d_{\mathrm{safe}}$ ($b_{ia} = b_{ja} = 0$) between a moving robot i and a stationary target-holding neighbor j ($\boldsymbol{e}_j = \boldsymbol{0}, \boldsymbol{u}_j^{\mathrm{nom}} = \boldsymbol{0}$), with unit contact normal $\boldsymbol{n}_{ij} = (\boldsymbol{p}_i - \boldsymbol{p}_j)/d_{\mathrm{safe}}$. Suppose the GNN residual policy (29) generates a lateral bypass nominal velocity for robot i with strictly positive normal intrusion $\delta_n := -\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_i^{\mathrm{nom}} > 0$ (unsaturated):*

1) ***Artificial freezing under static equal splitting*** *($B = 0, s_{ia} = s_{ja} \equiv 0$): The decoupled constraints reduce to $-2d_{\mathrm{safe}}\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_i \le 0$ and $2d_{\mathrm{safe}}\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_j \le 0$. Since $\boldsymbol{u}_j^{\mathrm{nom}} = \boldsymbol{0}$ is feasible for node j, $\boldsymbol{u}_j^{\mathrm{stat}} = \boldsymbol{0}$ (stationary neighbor remains frozen) while robot i's normal component is truncated to zero ($-\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_i^{\mathrm{stat}} = 0$).*
2) ***Quota transfer and cooperative yielding under closed-neighborhood dynamic allocation*** *($B \ge 1$): In the two-node contact block subproblem of (15) starting from zero quotas, the unique optimal pairwise quota transfer $t_{ij}^{\star}$, normal advance speed $-\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_i^{\star}$, and outward yielding speed $-\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_j^{\star}$ admit the strictly positive closed-form solution:*

$$
\begin{aligned}
|t_{ij}^{\star}| &= \frac{\eta d_{\mathrm{safe}}}{\eta + 4d_{\mathrm{safe}}^2}\delta_n > 0, \\
-\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_i^{\star} &= -\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_j^{\star} = \frac{\eta}{2(\eta + 4d_{\mathrm{safe}}^2)}\delta_n > 0.
\end{aligned}
\tag{31}
$$

Proof. At $d_{ij} = d_{\mathrm{safe}}$, $\boldsymbol{a}_{ia} = -2d_{\mathrm{safe}}\boldsymbol{n}_{ij}, \boldsymbol{a}_{ja} = 2d_{\mathrm{safe}}\boldsymbol{n}_{ij}, b_{ia} = b_{ja} = 0$. Decompose velocities into normal components $u_{q,n} = -\boldsymbol{n}_{ij}^{\top}\boldsymbol{u}_q$ (positive outward from i toward j) and tangential components $\boldsymbol{u}_{q,\tau} = (\boldsymbol{I}_2 - \boldsymbol{n}_{ij}\boldsymbol{n}_{ij}^{\top})\boldsymbol{u}_q$. Tangential components are unconstrained, so $\boldsymbol{u}_{i,\tau}^{\star} = \boldsymbol{u}_{i,\tau}^{\mathrm{nom}}, \boldsymbol{u}_{j,\tau}^{\star} = \boldsymbol{0}$. Along the normal direction: (1) For $B = 0, s_{ia} = s_{ja} = 0$, minimizing $\frac{1}{2}(u_{i,n} - \delta_n)^2$ subject to $2d_{\mathrm{safe}}u_{i,n} \le 0$ and $\frac{1}{2}u_{j,n}^2$ subject to $-2d_{\mathrm{safe}}u_{j,n} \le 0$ yields $u_{i,n}^{\mathrm{stat}} = u_{j,n}^{\mathrm{stat}} = 0$. (2) In the block QP (15), the normal constraints are $2d_{\mathrm{safe}}u_{i,n} + t_{ij} \le 0$ and $-2d_{\mathrm{safe}}u_{j,n} - t_{ij} \le 0$ with strictly convex cost $\frac{1}{2}(u_{i,n} - \delta_n)^2 + \frac{1}{2}u_{j,n}^2 + \frac{1}{\eta}t_{ij}^2$. Since the unconstrained minimum $(\delta_n, 0, 0)$ violates $2d_{\mathrm{safe}}u_{i,n} + t_{ij} \le 0$, both constraints are active at optimum: $t_{ij} = -2d_{\mathrm{safe}}u_{i,n} = -2d_{\mathrm{safe}}u_{j,n}$ and $u_{i,n}^{\star} = u_{j,n}^{\star} =: v_n$. Minimizing $f(v_n) = \frac{1}{2}(v_n - \delta_n)^2 + \frac{1}{2}v_n^2 + \frac{4d_{\mathrm{safe}}^2}{\eta}v_n^2$ gives $(2 + \frac{8d_{\mathrm{safe}}^2}{\eta})v_n = \delta_n$, yielding (31). □

Proposition 3 clarifies the complementary division of labor between the learned nominal policy and the dynamic quota filter: the policy network only needs to generate a lateral bypass component $\delta_n > 0$ for blocked moving robots, while the closed-neighborhood block QP automatically converts this normal intrusion into minimum-norm yielding motion of stationary neighbors via quota transfer $t_{ij}^{\star}$. Equation (31) also shows that the normal filter modification is $u_{i,n}^{\star} - \delta_n = -\frac{\eta + 8d_{\mathrm{safe}}^2}{2(\eta + 4d_{\mathrm{safe}}^2)}\delta_n \ne 0$: penalizing filter intervention in the policy loss suppresses $\delta_n > 0$ and weakens the quota transfer $|t_{ij}^{\star}|$ in (31), whose closed-loop impact is evaluated across both primary and matched factorial benchmarks in Section V.

C. One-Hop Closed-Neighborhood Completion Envelope and Goal Equilibrium Recovery

In (29), the tangential and radial residual terms scale with $\boldsymbol{e}_{i,k}$ and vanish as $\boldsymbol{e}_{i,k} \to \boldsymbol{0}$, whereas the pairwise repulsive bias $\boldsymbol{b}_{i,k}^{\psi} = \frac{A_{\psi}U}{\max(|\mathcal{N}_{i,k}|,1)}\sum_{j \in \mathcal{N}_{i,k}}\psi_{ij,k}^{\theta}(\boldsymbol{p}_{i,k} - \boldsymbol{p}_{j,k})/d_{ij,k}$ is a zero-order term of unit direction vectors. On a tightly packed goal lattice with spacing 1.001 m (1 mm above $d_{\mathrm{safe}} = 1.000$ m), near-active contact constraints maintain $\|\boldsymbol{\mu}_{i,k-1}\|_2 > 0$ and $\gamma_{i,k}^{\mathrm{gate}} \approx 1$, causing $\boldsymbol{b}_{i,k}^{\psi}$ to balance $-\boldsymbol{e}_{i,k}$ at $\|\boldsymbol{e}_{i,k}\|_2 \in [0.015, 0.05]$ m (terminal steady-state hovering).

To keep residuals active whenever any one-hop neighbor has not yet reached its goal while enforcing unconditional first-order vanishing once the entire one-hop closed neighborhood enters the goal ball, we define the *One-Hop Closed-Neighborhood Completion Envelope*:

$$
\begin{aligned}
\rho_{i,k}(\boldsymbol{p}_k) &= \tanh\left(\frac{\max_{j \in \mathcal{N}_{i,k}^{+}}\|\boldsymbol{p}_{j,k} - \boldsymbol{g}_j\|_2}{\varepsilon_g}\right), \\
\gamma_{i,k}^{\mathrm{eff}} &= \gamma_{i,k}^{\mathrm{gate}} \cdot \rho_{i,k}(\boldsymbol{p}_k),
\end{aligned}
\tag{32}
$$

with envelope scale $\varepsilon_g = 0.05$ m ($5\times$ the arrival tolerance 0.01 m).

Proposition 4 (First-Order Residual Vanishing and Goal Equilibrium Recovery)**.** *Under the completion envelope* (32) *in* (29)*:*

1) ***Residual preservation during cooperative transit:*** *Whenever there exists $j \in \mathcal{N}^+_{i,k}$ with $\|\boldsymbol{e}_{j,k}\|_2 \geq 3\varepsilon_g = 0.15$ m, even if $\boldsymbol{e}_{i,k} = \mathbf{0}$, we have $\rho_{i,k}(\boldsymbol{p}_k) \geq \tanh(3) > 0.995$.*
2) ***First-order suppression near the goal and equilibrium restoration:*** *Let $e^{\max}_{i,k} := \max_{j\in\mathcal{N}^+_{i,k}} \|\boldsymbol{e}_{j,k}\|_2 \leq \|\boldsymbol{e}_k\|_2$. Since $\tanh(z) \leq z$ $(\forall z \geq 0)$, $\gamma^{\text{gate}}_{i,k} \in [0,1)$, and $|\omega^\theta_{i,k}|, |r^\theta_{i,k}|, |\psi^\theta_{ij,k}| \leq 1$,*

$$\begin{aligned}\|\gamma^{\text{eff}}_{i,k}\boldsymbol{r}^\theta_{i,k}(\boldsymbol{p}_k)\|_2 &\leq \frac{e^{\max}_{i,k}}{\varepsilon_g}\Big[(A_{\text{rot}} + A_{\text{rad}})\|\boldsymbol{e}_{i,k}\|_2 + A_\psi U\Big] \\ &= \mathcal{O}(\|\boldsymbol{e}_k\|_2).\end{aligned} \tag{33}$$

In particular, at $\boldsymbol{p}_k = \boldsymbol{g}$, $\rho_{i,k}(\boldsymbol{g}) = 0$ and $\boldsymbol{u}^{\text{nom}}_{i,k}(\boldsymbol{g};\theta) = \mathbf{0}$ regardless of $\boldsymbol{\mu}_{i,k-1}$.

Proof. Claim (1) follows from monotonicity of $\tanh(\cdot)$ since $\tanh(3) \approx 0.99505 > 0.995$. Claim (2) follows from $0 \leq \gamma^{\text{eff}}_{i,k} \leq \rho_{i,k}(\boldsymbol{p}_k) \leq e^{\max}_{i,k}/\varepsilon_g$ and the triangle inequality $\|\boldsymbol{r}^\theta_{i,k}(\boldsymbol{p}_k)\|_2 \leq (A_{\text{rot}} + A_{\text{rad}})\|\boldsymbol{e}_{i,k}\|_2 + A_\psi U$. Setting $\boldsymbol{e}_k = \mathbf{0}$ yields $\boldsymbol{u}^{\text{nom}}_{i,k}(\boldsymbol{g};\theta) = \mathbf{0}$. □

D. Small-Scale Short-Horizon MAPPO Training and Zero-Shot Cross-Scale Deployment

Because the B-round block QP filter L involves nonsmooth active-set switches and local settlements, backpropagating long-horizon policy gradients through the QP solver suffers from gradient singularities at multi-contact corners. We adopt a Centralized Training with Decentralized Execution (CTDE) MAPPO protocol [22] that combines small-scale short-horizon training ($N \in \{6,\dots,10\}, B = 2$, validation selection on $N = 8$) with large-scale multi-round zero-shot deployment ($N \in \{12, 20, 25\}, B = 8$), as summarized in Table I. Training tasks are sampled on dense triangular lattices (spacing $1.001 \sim 1.12$ m, $\lambda_{\text{req}}/\lambda_2(\boldsymbol{p}_0) \in [0.55, 0.90]$, $1 \sim 3$ pair swaps, 3-cycle rotations, and full permutations with random global rotation). The shared team reward at step k is:

$$\begin{aligned}R_k = r_{\text{prog}}&\frac{\sum_{i=1}^N \big(\|\boldsymbol{e}_{i,k}\|_2 - \|\boldsymbol{e}_{i,k+1}\|_2\big)}{D_0} - r_{\text{time}} \\ &+ r_{\text{succ}}\mathbb{I}\Big(\max_{i\in\mathcal{V}}\|\boldsymbol{e}_{i,k+1}\|_2 \leq \varepsilon_{\text{succ}}\Big) \\ &- \frac{r_{\text{stall}}}{N}\sum_{i=1}^N \tanh\Big(\frac{\|\boldsymbol{e}_{i,k}\|_2}{\varepsilon_p}\Big)\exp\Big(-\frac{\|\boldsymbol{u}_{i,k}\|_2^2}{\varepsilon_u^2}\Big) \\ &\times \ln\big(1 + \|\boldsymbol{\mu}_{i,k}\|_2^2\big),\end{aligned} \tag{34}$$

where $D_0 = \sum_{i=1}^N \|\boldsymbol{e}_{i,0}\|_2$, $r_{\text{prog}} = 1.0$, $r_{\text{time}} = 10^{-3}$, $\varepsilon_{\text{succ}} = 0.01$ m, $r_{\text{succ}} = 1.0$, and the fourth term penalizes dual-blocked stagnation ($r_{\text{stall}} = 0.02, \varepsilon_p = 0.10$ m, $\varepsilon_u = 0.05$ m/s). Policy parameters θ are updated via GAE ($\gamma = 0.995, \lambda_{\text{GAE}} = 0.95$) and clipped PPO ($\epsilon_{\text{clip}} = 0.20$, target KL 0.03).

TABLE I
EGP-NPA TRAINING AND ZERO-SHOT DISTRIBUTED EXECUTION (ALGORITHM 1)

Phase A: Small-Scale Short-Horizon CTDE Training ($N \in \{6,\dots,10\}$, $B_{\text{train}} = 2$)

1. Initialize 2-layer equivariant GNN actor θ and centralized GNN critic ϕ;
2. **for** iteration $m = 1, 2, \dots, M_{\max}$ **do**
3. Sample dense triangular-lattice swap/permutation tasks ($N \in \{6,\dots,10\}$, random rotation), set $\boldsymbol{\varpi}_{-1} = \mathbf{0}$;
4. **for** step $k = 0, 1, \dots, K_{\max} - 1$ **do**
5. Construct invariant features $(\boldsymbol{x}_{i,k}, \boldsymbol{e}^{\text{feat}}_{ij,k})$ and sample noisy nominal velocity $\boldsymbol{u}^{\text{nom}}_{i,k}$ via (27)–(29);
6. Solve single-robot refit QP (14) (recording shadow prices $\boldsymbol{\mu}_{i,k}$) and apply one-hop local settlement if needed;
7. Run $B_{\text{train}} = 2$ closed-neighborhood block NPA rounds (15)–(16) and ZOH certificate (Theorem 1);
8. Compute team reward R_k via (34) and store transition in rollout buffer;
9. **end for**
10. Compute GAE advantages, update (θ, ϕ) via clipped PPO, and save best checkpoint $\theta^\star$ on $N = 8$ validation set.
11. **end for**

Phase B: Zero-Shot Cross-Scale Decentralized Execution (Arbitrary N, $\theta = \theta^\star$, $B_{\text{eval}} = 8$)

12. At each step t_k, each robot $i \in \mathcal{V}$ evaluates completion envelope $\rho_{i,k}(\boldsymbol{p}_k) = \tanh(\max_{j\in\mathcal{N}^+_{i,k}} \|\boldsymbol{e}_{j,k}\|_2/\varepsilon_g)$;
13. Compute deterministic GNN actions $(\omega^{\theta^\star}_{i,k}, r^{\theta^\star}_{i,k}, \psi^{\theta^\star}_{ij,k})$ and synthesize $\boldsymbol{u}^{\text{nom}}_{i,k}(\boldsymbol{p}_k;\theta^\star)$ via (29);
14. Execute single-robot refit (14), one-hop settlement, and $B_{\text{eval}} = 8$ block NPA rounds (15)–(16) to apply $\boldsymbol{u}_{i,k}$.

V. Closed-Loop Numerical Validation, Mechanism Decomposition, and Virtual-Physical Verification

A. Experimental Setup, Baselines, and Evaluation Metrics

We evaluate EGP-NPA across training-scale ($N = 8$) and zero-shot cross-scale transfer ($N \in \{12, 20, 25\}$) dense triangular-lattice swap and random full-permutation benchmarks. All experiments share identical physical and safety parameters: $d_{\text{safe}} = 1.000$ m, $U = 0.50$ m/s, $B_{\text{ws}} = 4.0$ m, $R_c = 10.0$ m, $R_{\text{sense}} = 2.50$ m, $\alpha_c = \alpha_s = \alpha_x = 2.0$ s^{-1}, $\Delta t = 0.05$ s, and horizon $T_{\max} = 60.0$ s. Initial configurations are tightly packed triangular lattices with spacing 1.001 m (1.0 mm initial clearance). Fig. 2 shows the MAPPO training return and deterministic validation arrival rate ($N = 8$, 36 validation instances) over 9.6×10^5 environment steps: the zero-residual initialization (myopic proportional attraction) achieves only 19.4% arrival, whereas EGP-NPA converges to 100% validation arrival.

The primary benchmark comprises 96 independent instances:

1) **Training-scale benchmark ($N = 8$, 64 instances):** 20 local pair-swap instances ($1 \sim 3$ adjacent pairs swap goals while remaining nodes stay at their goals) and 12 random full-permutation instances, evaluated under both nominal connectivity threshold $f = \lambda_{\text{req}}/\lambda_2(\boldsymbol{p}_0) = 0.60$ (32 instances) and tight connectivity threshold $f = 0.90$ (32 instances).
2) **Zero-shot cross-scale transfer benchmark ($N \in \{12, 20, 25\}$, 32 instances):** $N = 12$ (8 local swaps + 4 full permutations), $N = 20$ (8 local swaps + 4 full

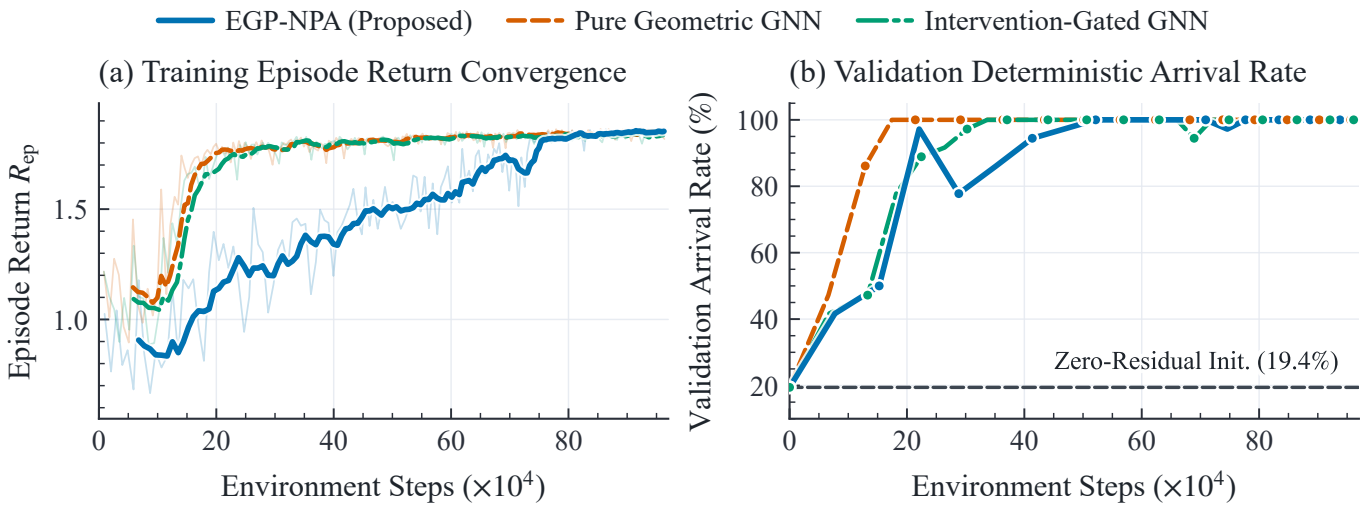


Fig. 2. MAPPO training evolution over 9.6×10^5 environment steps. (a) Episode return R_{ep} convergence (faint lines: raw episode returns; bold lines: moving average); (b) deterministic arrival rate on the $N = 8$ validation benchmark (36 swap and permutation instances).

TABLE II
CLOSED-LOOP COMPARISON ON THE TRAINING-SCALE BENCHMARK ($N = 8$, 32 INSTANCES EACH AT $f = 0.60$ AND $f = 0.90$) AGAINST FOUR SAFE RL METHODS AND CLASSICAL BASELINES

Control Method	$f = 0.60$ Arrival	Median Time	$f = 0.90$ Arrival	Median Time	Step Cuts	Audit Viol.
EGP-NPA (Proposed)	**32/32 (100%)**	**7.32 s**	**32/32 (100%)**	**7.12 s**	**0**	**0**
Shield-Consistent Safe MARL (`Proj-MAPPO + NPA` [16], [17], $B = 8$)	32/32 (100%)	9.53 s	32/32 (100%)	10.73 s	0	0
Static-Split CBF Shielded GNN (`Static-Split GNN` [4], [18], $B = 0$)	5/32 (15.6%)	13.30 s	6/32 (18.8%)	13.88 s	431 725	0
Discrete GCBF PPO (`DGPPO` [19], Official Task-Adapted)	0/32† (0%)	—	0/32† (0%)	—	—	76 800
Primal-Dual Safe MARL (`MAPPO-Lagrangian` [15], Unshielded)	0/32‡ (0%)	4.15 s‡	0/32‡ (0%)	4.15 s‡	—	7 047
Circulation Heuristic [13] + Dynamic NPA ($B = 8$)	21/32 (65.6%)	32.50 s	18/32 (56.3%)	30.50 s	0	0
Myopic CBF Filter [4], [7] + Dynamic NPA ($B = 8$)	17/32 (53.1%)	36.35 s	16/32 (50.0%)	31.80 s	0	0
Circulation Heuristic + Static Equal Split [4], [18] ($B = 0$)	2/32 (6.3%)	25.10 s	—	—	72 156	0

Note: †Task-adapted official `DGPPO` incurs collision violations on all 64 runs across both training seeds (76 800 and 76 789 violated ZOH segments); ‡`MAPPO-Lagrangian` holds the goal tolerance for 1.0 s in 64/64 runs (median 4.15 s) but violates collision safety on all 64 runs (7 047 violated ZOH segments, median $\min d_{ij} = 0.263$ m $< d_{\text{safe}} = 1.000$ m), yielding 0/32 safe arrivals for both unshielded methods.

permutations), and $N = 25$ (8 local swaps) at $f = 0.60$, directly deploying the frozen small-scale checkpoint $\theta^\star$ without retraining.

We compare EGP-NPA against **four Safe MARL comparison methods** and **three classical analytical/heuristic CBF baselines**:

1) **Shield-Consistent / Action-Projected Safe MARL (`Proj-MAPPO + NPA` [16], [17]):** Represents recent predictive/adaptive safety-filter RL frameworks that penalize filter intervention during training. It shares the identical 2-layer equivariant GNN, completion envelope $\rho_{i,k}$, and $B = 8$ closed-neighborhood NPA shield, but adds the shield-projection consistency loss $\mathcal{L}_{\text{proj}}(\theta) = \lambda_{\text{proj}} \frac{1}{N} \sum_{i=1}^{N} \|\Pi_{\mathcal{U}_i}(\boldsymbol{u}_{i,k}^{\text{nom}}(\theta)) - \text{sg}(\boldsymbol{u}_{i,k})\|_2^2$ ($\lambda_{\text{proj}} = 1.0$ calibrated by gradient norm ratio) to align nominal actions with filtered safe velocities;
2) **Static-Split CBF Shielded GNN (`Static-Split GNN` [4], [18], $B = 0$):** Combines the equivariant GNN policy and completion envelope $\rho_{i,k}$ with decentralized single-agent static equal-responsibility CBF-QP filtering ($B = 0$, fixing $s_{ia,k} \equiv 0$ without closed-neighborhood dynamic quota transfer t_e);
3) **Discrete GCBF Proximal Policy Optimization (`DGPPO` [19], Official Task-Adapted Implementation):** Uses the official public `DGPPO` implementation of Zhang et al. [19] (graph-transformer policy, task value network V^l, decomposed constraint value network V^h, and discrete barrier advantage update), adapted to our single-integrator kinematics and the three coupled scalar constraints (pairwise collision, global scalar connectivity, and workspace bounds, retaining the upstream cost-margin transformation), trained for 4.8×10^5 environment transitions (500-step rollout horizon across two seeds over $N \in \{6, \ldots, 10\}$) and executed directly;
4) **Primal-Dual Constrained Safe MARL (`MAPPO-Lagrangian` [15]):** Represents model-free Constrained Markov Decision Process (CMDP) safe MARL without a hard coupled QP shield, training

the identical 2-layer equivariant GNN actor, task critic, and cost critic with an adaptive dual-ascent Lagrangian multiplier $\lambda_{\text{lag}} \geq 0$ penalizing collision and connectivity violations;

5) **Classical Analytical and Heuristic CBF Baselines:** Tangential Circulation Heuristic of Gonçalves et al. [13] ($\omega_i \equiv 1, r_i \equiv 0, \psi_{ij} \equiv 0$ with $B = 8$), Myopic Proportional Attraction CBF filtering of Ames et al. [4], [7] ($\boldsymbol{u}_i^{\text{nom}} = -\boldsymbol{e}_i$ with $B = 8$), and Circulation Heuristic with static quota splitting ($B = 0$).

Safe arrival requires $\max_{i\in\mathcal{V}} \|\boldsymbol{p}_i(t) - \boldsymbol{g}_i\|_2 \leq 0.01$ m maintained continuously for 1.0 s with 0 rational safety audit violations across all ZOH segments (Theorem 1). Statistical significance is evaluated via two-sided McNemar exact binomial tests on discordant pairs and median paired arrival-time differences Δt on common arrivals.

B. Closed-Loop State Evolution on Dense Interior Swap and Full Permutation

Figs. 3 and 4 compare the spatial trajectories, communication topologies, and closed-loop time responses on a representative $N = 20$ interior pair-swap task (2 interior robots swap positions while 18 outer robots start at their goals).

Three physical mechanisms are evident from Figs. 3 and 4:

1) **Chiral lateral bypass and stationary neighbor yielding–return loop:** Under Static-Split CBF Shielded GNN (`Static-Split GNN`, $B = 0$, Fig. 3(d)), outer stationary target-holding neighbors keep $s_{ja} = 0$ and refuse to yield collision quotas, truncating the swapping pair's lateral velocity and driving RMS speed to zero (purple dotted curve in Fig. 4(d)). Under Shield-Consistent Safe MARL (`Proj-MAPPO + NPA`, Fig. 3(c)), penalizing normal filter intervention weakens the normal intrusion $\delta_n > 0$, trapping the swapping pair in a local livelock at 1.01 m (or arriving slowly at 16.90 s in Fig. 4(a)). Under EGP-NPA (Fig. 3(a)), the equivariant GNN generates chiral tangential thrust for the swapping pair, and closed-neighborhood dynamic quota transfer converts this lateral push into compact outward yielding loops (bluish-green trajectories) of adjacent stationary neighbors, which smoothly return to their goals once the swapping pair passes.
2) **Elimination of terminal steady-state hovering by the completion envelope:** In Fig. 4(a) and (d), the ablation without the completion envelope (orange dashed curve) resolves the initial deadlock within 8 s but remains trapped in a limit cycle at $\max_i \|\boldsymbol{e}_i\|_2 \in [0.015, 0.05]$ m with persistent RMS speed $0.01 \sim 0.04$ m/s due to nonvanishing zero-order repulsive bias $\boldsymbol{b}_{i,k}^{\psi}$. With the one-hop completion envelope $\rho_{i,k}$ (32), residuals vanish to first order inside $\varepsilon_g = 0.05$ m (Proposition 4), driving the error cleanly below 0.01 m and bringing control speed to zero.
3) **Strict continuous-time safety invariance:** Throughout the maneuver (Fig. 4(b)–(c)), $\lambda_2(\boldsymbol{L}_{\text{p}}(t)) \geq 15.3 > \lambda_{\text{req}} = 9.69$ and $\min_{i<j} d_{ij}(t) \geq d_{\text{safe}} = 1.000$ m with zero violations.

C. Comparison With External Baselines on Training and Cross-Scale Benchmarks

Tables II and III summarize the statistical results across the 64 training-scale instances ($N = 8$) and 32 zero-shot cross-scale instances ($N \in \{12, 20, 25\}$), establishing four key findings:

1) **Unshielded safe MARL (`DGPPO` [19] and `MAPPO-Lagrangian` [15]) cannot maintain hard collision safety on millimeter-clearance lattices:** On tightly packed lattices (1.0 mm initial clearance), task-adapted official `DGPPO` incurs collision violations on all 96 runs across both training seeds (115 200 and 115 186 violated segments, 0% safe arrival); similarly, despite dual ascent raising λ_{lag} to 2.45 and reaching the goal vicinity in $3.92 \sim 4.15$ s, `MAPPO-Lagrangian` incurs 10 971 violated ZOH segments across all 96 runs (0% safe arrival), confirming the necessity of the hard coupled NPA shield in Proposition 2.
2) **Static-split CBF shielded GNN (`Static-Split GNN` [4], [18], $B = 0$) is severely obstructed under multi-layer stationary wrapping:** Sharing the exact same GNN policy weights and completion envelope as EGP-NPA but using static equal responsibility splitting ($B = 0$), `Static-Split GNN` achieves only 17.2% (11/64) safe arrival on $N = 8$ (431 725 step cuts) and drops to 3.1% (1/32, discordant pairs $31 : 0, p = 9.3 \times 10^{-10}$) on $N \in \{12, 20, 25\}$ (across all 96 runs: 12 safe arrivals, 25 zero-speed stalls, 37 step-halving certification stops, 14 call-budget stops, and 8 timeouts), verifying Proposition 3 Claim 1.
3) **Shield-consistent safe MARL (`Proj-MAPPO + NPA` [16], [17]) slows cooperative yielding:** While achieving 64/64 arrival on $N = 8$ (2.88 s slower in median time), the validation-selected checkpoint reaches 53.1% (17/32, discordant pairs $15 : 0, p = 6.1 \times 10^{-5}$, −3.85 s slower) on $N \in \{12, 20, 25\}$, consistent with Proposition 3 Claim 2 that penalizing filter intervention weakens the normal intrusion drive $\delta_n > 0$.
4) **Superiority over classical analytical/heuristic baselines:** As N scales to 25, Myopic CBF filtering and Circulation Heuristic drop to 15.6% ($27 : 0, p = 1.5 \times 10^{-8}$) and 40.6% ($19 : 0, p = 3.8 \times 10^{-6}$), whereas EGP-NPA achieves 96/96 (100%) safe arrival with 0 step cuts and 0 audit violations.

D. Internal Module Ablation, Velocity Decomposition, and Holdout Factorial Analysis

Table IV reports: (A) cross-scale ablations removing the completion envelope ($\rho_{i,k} \equiv 1$) and comparing training checkpoints; (B) deadlock-breaking velocity decomposition on $N = 20$ ($t \in [0.05, 1.00]$ s) under identical EGP-NPA nominal inputs comparing dynamic allocation ($B = 8$) vs. static splitting ($B = 0$); and (C) controlled single-variable execution-time interventions holding the trained weights $\theta^\star$ strictly fixed (16 cross-scale instances; Fig. 5(b)).

Table IV confirms three causal mechanisms:

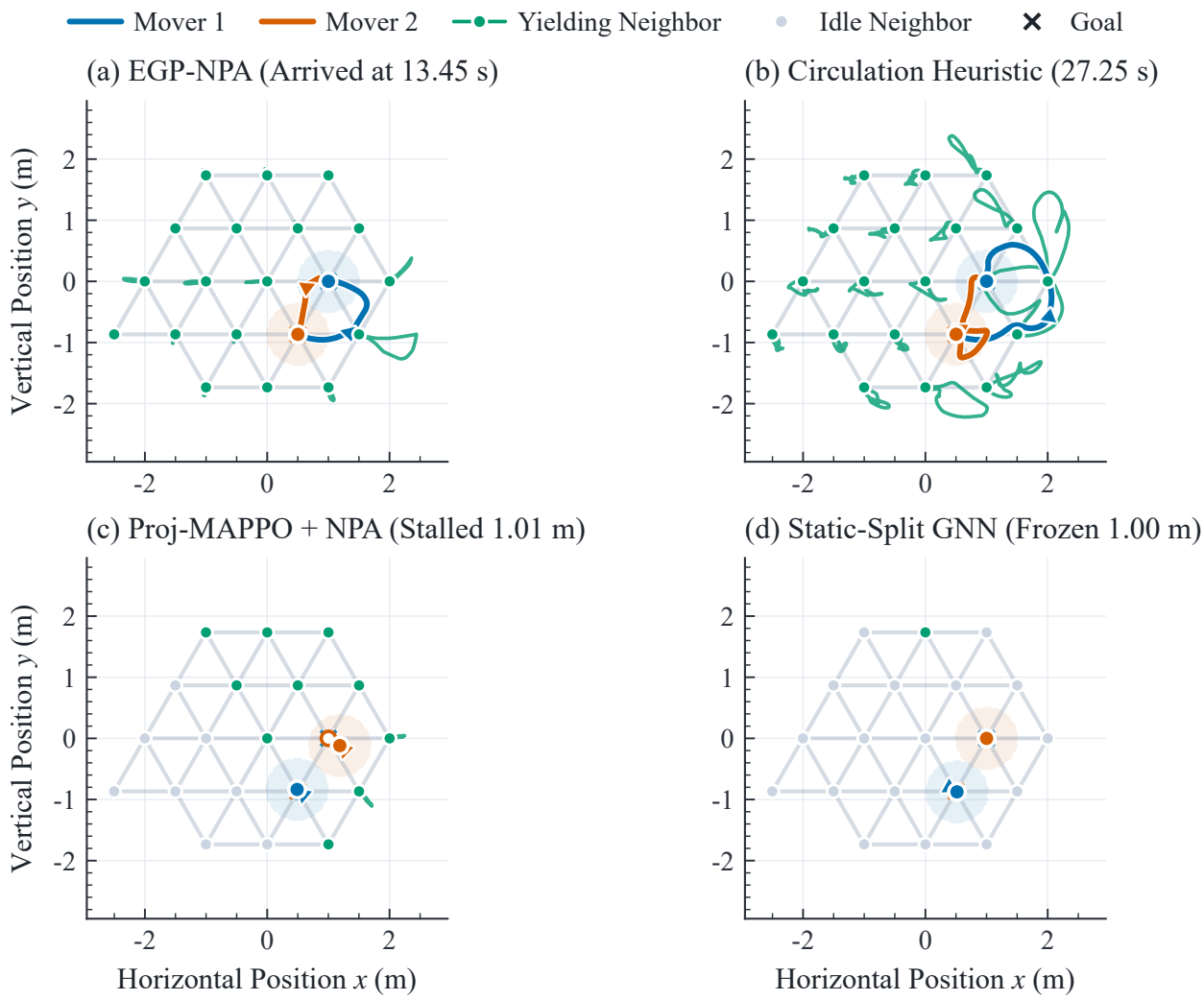


Fig. 3. Spatial trajectories and communication topologies on the $N = 20$ dense triangular-lattice interior pair-swap task. (a) EGP-NPA drives the swapping pair along smooth tangential bypasses while unfreezing outer stationary neighbors (bluish-green loops) via dynamic quota transfer, arriving at 13.45 s; (b) Circulation Heuristic [13] triggers large-scale rotational conflicts among outer neighbors (27.25 s); (c) Shield-Consistent Safe MARL (`Proj-MAPPO + NPA` [16], [17]) stalls in a livelock at 1.01 m because penalizing normal filter modification weakens the yielding drive $\delta_n > 0$; (d) Static-Split CBF Shielded GNN (`Static-Split GNN` [4], [18], $B = 0$) freezes at 1.00 m due to static quota splitting.

1) **Completion envelope necessity (Part A):** Removing $\rho_{i,k}$ while holding $\theta^\star$ fixed induces 7 terminal hovering failures on $N = 20$ ($7 : 0, p = 0.0156$, $+2.40$ s slower), directly verifying Proposition 4. Note that the pure-geometric and intervention-gated rows in Part A reflect independently trained single-seed checkpoints (whose differences combine configuration effects with inter-seed training variance), whereas Parts B and C isolate strict causal effects under fixed weights $\theta^\star$.
2) **Division of labor between learned lateral intrusion and dynamic quota yielding (Part B):** During deadlock breaking, the GNN outputs a lateral nominal speed of 0.8887 m/s for the swapping pair (with 0.000 action saturation across all runs), while stationary neighbors output virtually zero outward nominal residual (-0.0105 m/s). Under $B = 8$ dynamic allocation, block QPs transfer collision quotas to drive stationary neighbors outward at $+0.0915$ m/s, allowing swapping nodes to advance laterally at 0.2003 m/s; under static splitting ($B = 0$), these speeds collapse to $+0.0013$ m/s and 0.0004 m/s, confirming Proposition 3.
3) **Execution-time gate simplification and feature knockout (Part C):** Holding $\theta^\star$ fixed, setting $\gamma_{i,k}^{\text{gate}} \equiv 1$ (retaining only $\rho_{i,k}$) achieves $16/16$ arrival with trajectories indistinguishable from $\gamma_{i,k}^{\text{gate},\mu}$ (Fig. 5(b)). Across all 32 cross-scale instances, zeroing only the 4 collision shadow-price columns reduces arrival to $31/32$ and slows down 25 of 31 common arrivals (two-sided sign test $p = 8.8\times10^{-4}$; $+1.18$ s on the 16-instance subset), proving active utilization of refit shadow prices $\boldsymbol{\mu}_{i,k-1}$, whereas zeroing collision ledgers or replacing the 3 Laplacian eigenvector columns with their median constant yields $32/32$ arrival with 0.00 s median difference.

On 48 additional independent holdout instances (32 at $N = 8$ and 16 at $N \in \{12, 20, 25\}$) plus 2 exact mirror-symmetric head-on swap benchmarks, we further conduct a

TABLE III
ZERO-SHOT CROSS-SCALE TRANSFER BENCHMARK ($N \in \{12, 20, 25\}$, 32 INSTANCES TOTAL) AGAINST FOUR SAFE RL METHODS AND CLASSICAL BASELINES

Control Method	$N = 12$ (12)	$N = 20$ (12)	$N = 25$ (8)	Total Arrival (32)	Discordant (p-val)	Median Δt	Step Cuts
EGP-NPA (Proposed)	**12/12 (6.52 s)**	**12/12 (10.55 s)**	**8/8 (10.43 s)**	**32/32 (100%)**	Baseline	Baseline	**0**
Proj-MAPPO + NPA [16], [17] ($B = 8$)	9/12 (17.25 s)	5/12 (17.70 s)	3/8 (12.30 s)	17/32 (53.1%)	$15 : 0$ (6.1×10^{-5})	-3.85 s	0
Static-Split GNN [4], [18] ($B = 0$)	0/12 (—)	0/12 (—)	1/8 (19.85 s)	1/32 (3.1%)	$31 : 0$ (9.3×10^{-10})	-12.95 s	447 022
DGPPO [19] (Official Task-Adapted)	0/12† (—)	0/12† (—)	0/8† (—)	0/32† (0%)	$32 : 0$ (4.7×10^{-10})	—	38 400†
MAPPO-Lagrangian [15] (Unshielded)	0/12‡ (4.05 s)	0/12‡ (3.95 s)	0/8‡ (3.92 s)	0/32‡ (0%)	$32 : 0$ (4.7×10^{-10})	—	3 924‡
Circulation Heuristic [13]	7/12 (14.40 s)	4/12 (24.53 s)	2/8 (38.22 s)	13/32 (40.6%)	$19 : 0$ (3.8×10^{-6})	-12.35 s	14 198
Myopic CBF Filter [4], [7]	4/12 (34.30 s)	1/12 (55.50 s)	0/8 (—)	5/32 (15.6%)	$27 : 0$ (1.5×10^{-8})	-26.70 s	0

Note: All five shielded methods incur 0 rational audit violations; unshielded †`DGPPO` and ‡`MAPPO-Lagrangian` violate collision safety on all 32 cross-scale runs (38 400 and 3 924 violated ZOH segments recorded in the last column), yielding $0/32$ safe arrivals.

TABLE IV
INTERNAL MODULE CROSS-SCALE ABLATION (PART A, 32 INSTANCES), DEADLOCK-BREAKING VELOCITY DECOMPOSITION (PART B, $t \in [0.05, 1.00]$ s), AND FIXED-WEIGHT ($\theta^\star$) SINGLE-VARIABLE EXECUTION INTERVENTIONS (PART C, 16 INSTANCES)

A. Cross-Scale Module Ablation (32 Instances)	$N = 12$ (12)	$N = 20$ (12)	$N = 25$ (8)	Total Arrival (32)	Discordant (p-val) / Δt
EGP-NPA (Full Architecture)	12/12 (6.52 s)	12/12 (10.55 s)	8/8 (10.43 s)	**32/32 (100%)**	Baseline
w/o Completion Envelope (fixed $\theta^\star$, $\rho_{i,k} \equiv 1$)	12/12 (9.78 s)	5/12 (36.75 s)	8/8 (14.10 s)	25/32 (78.1%)	$7 : 0$ (0.0156) / -2.40 s
Pure Geometric GNN (trained w/o algebraic features)	12/12 (7.57 s)	7/12 (21.90 s)	3/8 (19.65 s)	22/32 (68.8%)	$10 : 0$ (0.0020) / -0.45 s
Intervention-Gated GNN ($\gamma_{i,k}^{\mathrm{gate,int}}$ checkpoint)	10/12 (11.20 s)	4/12 (30.08 s)	1/8 (6.80 s)	15/32 (46.9%)	$17 : 0$ (1.5×10^{-5}) / -0.50 s

B. Velocity Decomposition Under Identical EGP-NPA Input ($N = 20$ Swap)	Dynamic ($B = 8$)	Static ($B = 0$)
Swapping robots' mean tangential (lateral) nominal speed $\lvert\hat{\boldsymbol{e}}_i^\top \boldsymbol{J}^\top \boldsymbol{u}_i^{\mathrm{nom}}\rvert$	0.8887 m/s	0.8887 m/s
Swapping robots' mean tangential (lateral) executed speed $\lvert\hat{\boldsymbol{e}}_i^\top \boldsymbol{J}^\top \boldsymbol{u}_i\rvert$	**0.2003** m/s	0.0004 m/s
Outer 1-hop stationary neighbors' mean outward nominal residual speed	-0.0105 m/s	-0.0105 m/s
Outer 1-hop stationary neighbors' mean outward executed yielding speed	$+\mathbf{0.0915}$ m/s	$+0.0013$ m/s

C. Fixed-Weight ($\theta^\star$) Execution-Time Interventions (16 Instances)	$N = 12$ (6)	$N = 20$ (6)	$N = 25$ (4)	Arrival (16)	Paired Δt / Cuts
Native shadow-price gate $\gamma_{i,k}^{\mathrm{gate},\mu}$	6/6 (6.52 s)	6/6 (11.48 s)	4/4 (9.98 s)	16/16 (100%)	Baseline / 0
Stateless envelope-only gate ($\gamma_{i,k}^{\mathrm{gate}} \equiv 1$, retaining only $\rho_{i,k}$)	6/6 (6.40 s)	6/6 (10.03 s)	4/4 (9.73 s)	16/16 (100%)	-0.10 s / 0
Filter-intervention gate replacement ($\gamma_{i,k}^{\mathrm{gate,int}}$ at execution)	6/6 (6.82 s)	6/6 (9.35 s)	4/4 (10.43 s)	16/16 (100%)	$+0.33$ s / 0
Post-negotiation multiplier extraction with cross-step EMA	6/6 (7.03 s)	6/6 (12.48 s)	4/4 (9.73 s)	16/16 (100%)	$+0.03$ s / 0
In-distribution knockout of collision shadow prices (4 cols zeroed)	6/6 (7.63 s)	6/6 (13.38 s)	4/4 (10.50 s)	16/16 (100%)	$+\mathbf{1.18}$ s ($p = 0.0042$) / 0
In-distribution knockout of collision edge ledgers ($\varpi_{ij,1} = 0$)	6/6 (6.40 s)	6/6 (10.03 s)	4/4 (9.80 s)	16/16 (100%)	0.00 s ($p = 1.0$) / 0
Zeroing all 12 algebraic features (including constant reserve col.)	6/6 (9.85 s)	5/6 (9.80 s)	4/4 (13.03 s)	15/16 (93.8%)	$+1.30$ s ($p = 0.0074$) / 2 193

matched-budget two-seed 2×2 factorial experiment (4.8×10^5 environment steps, team vs. local critic, projection regularization $\lambda_{\mathrm{proj}} \in \{0, 1\}$, 8 checkpoints and 384 closed-loop runs with 0 step cuts and 0 audit violations): (1) the full-budget EGP-NPA checkpoint achieves $48/48$ (100%) on the holdout suite and $2/2$ on symmetric swaps, while the two half-budget retrained team-critic base checkpoints achieve $44/48$ and $46/48$ on the holdout suite (and $93/96$ and $92/96$ on a newly generated 96-instance external test suite: $64/64$ on $N = 8$ and $29/32, 28/32$ cross-scale); (2) enforcing $\mathrm{O}(2)$ reflection equivariance via odd–even feature symmetrization forces $\mathrm{mean}|\omega| = 0.0000$ on symmetric head-on swaps (100% deadlock) and drops holdout arrival from $48/48$ to $30/48$ ($62.5\%, p = 7.6 \times 10^{-6}$), confirming Proposition 1; (3) in matched-seed pairs, adding $\mathcal{L}_{\mathrm{proj}}$ ($\lambda_{\mathrm{proj}} = 1$) reduces filter intervention by $40\% \sim 70\%$ across all 8 pairs, changes cross-scale arrivals from $12/16, 14/16$ to $9/16, 11/16$ under the team critic (and from $4/16, 9/16$ to $8/16, 14/16$ under the

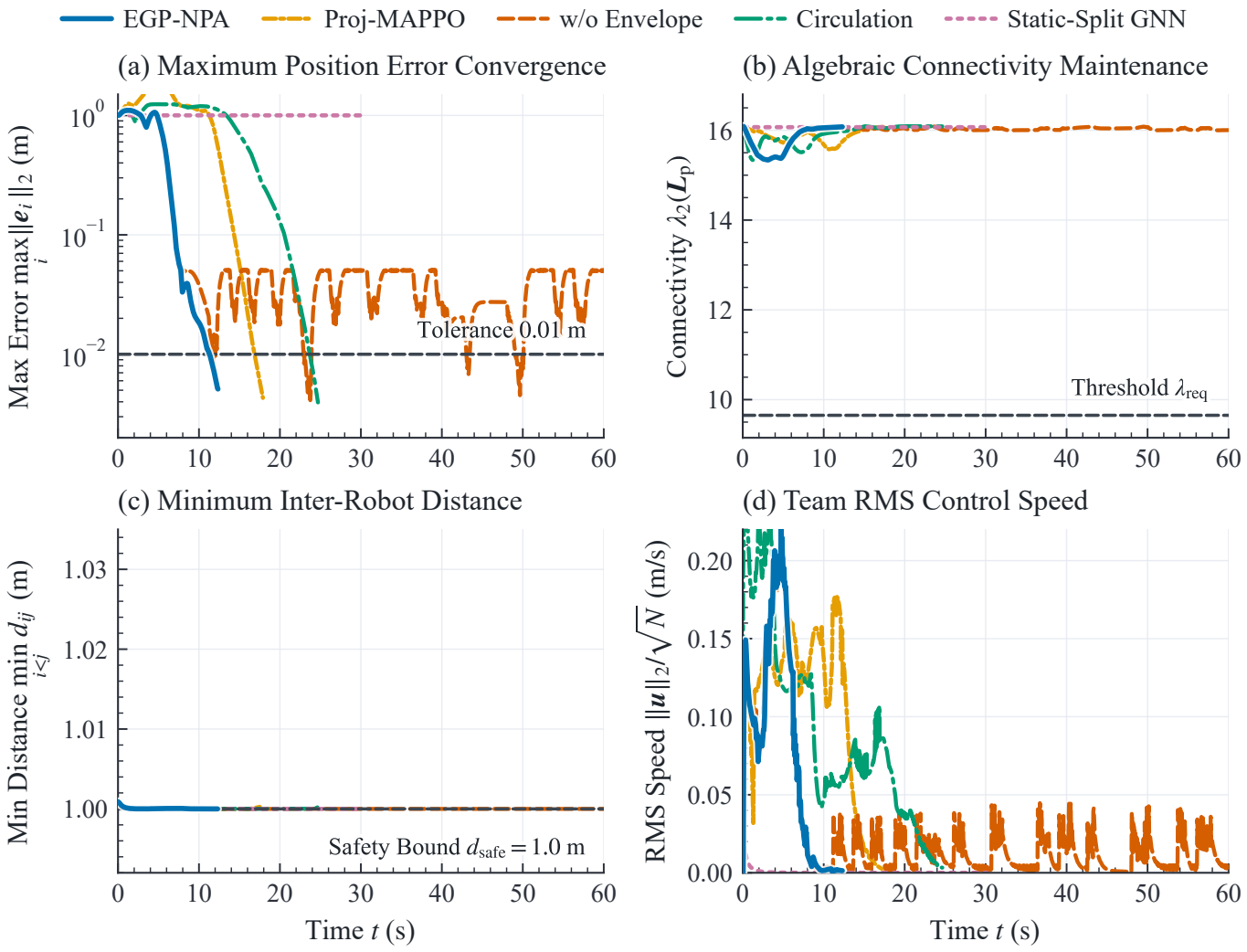


Fig. 4. Closed-loop time responses on the $N = 20$ interior swap task. (a) Maximum position error $\max_i \|e_i(t)\|_2$ (log scale); (b) algebraic connectivity $\lambda_2(L_p(t))$ vs. $\lambda_{req} = 9.69$; (c) minimum inter-robot distance $\min_{i<j} d_{ij}(t)$ vs. $d_{safe} = 1.0$ m; (d) team RMS control speed $\|u(t)\|_2/\sqrt{N}$.

local critic), and slows common arrivals by a median of $+2.00$ s across all 160 common pairs (116 slower vs. 43 faster, $p = 6.2 \times 10^{-9}$) while increasing 99th-percentile command jumps; and (4) replacing the global pooled critic with a local neighborhood critic delays convergence to 100% validation arrival by $16 \sim 24$ iterations ($+0.75$ s slower) since 2.50 m neighborhoods already cover 91.5% of the team at $N \leq 10$. Increasing evaluation rounds from $B = 2$ to $B = 8$ on $N = 8$ reduces median arrival time from 8.07 s to 7.42 s (32/32 arrival for both), within 0.33 s of exact centralized per-step QP ($B \to \infty$) owing to cross-step ledger persistence.

E. Zero-Shot Virtual-Physical Verification on Second-Order Chassis in MuJoCo

Finally, we evaluate zero-shot transfer of the frozen single-integrator EGP-NPA policy $\theta^\star$ and completion envelope $\rho_{i,k}$ when integrated with an online division-free 3-channel distributed spectral estimator (6 consensus rounds per step) and deployed on second-order omnidirectional robots in `MuJoCo 3.6.0` (200 Hz physics step $\delta t = 5$ ms, robot mass $m =$ 6.0 kg, body radius $r_{body} = 0.45$ m, velocity-loop time constant $\tau_v = 45$ ms, thrust limit $F_{max} = 22.0$ N, control period $\Delta t = 0.05$ s, observation delay 10 ms, packet delay $\tau_{comm} \in [5, 15]$ ms, packet drop rate 2%, and $B = 4$ block NPA rounds).

Theorem 3 (Critical Channel Pinch-Off Under Physical Margin Shrinkage)**.** *Consider N robots swapping positions on a triangular lattice with initial spacing $S > d_{safe}$, where outer neighbors j, m can open a maximum lateral aperture d_{jm}^{max}. Under physical safety margin shrinkage $d_{safe}^{phys} = d_{safe} + \Delta d_{phys}$ ($\mathcal{S}^{phys} = \{p : \min_{a<b} d_{ab} \geq d_{safe} + \Delta d_{phys}\}$):*

1) ***Static initial feasibility:*** *$p(0) \in \mathrm{int}(\mathcal{S}^{phys}) \iff \Delta d_{phys} < S - d_{safe}$.*
2) ***Three-body topological channel pinch-off:*** *For two counter-crossing robots i_1, i_2 passing simultaneously between j and m, maintaining pairwise distances $\geq d_{safe}^{phys}$ at the symmetric midpoint requires a contact rhombus of side length $\geq d_{safe}^{phys}$, demanding a minimum lateral aperture $d_{jm}^{req}(\Delta d_{phys}) = \sqrt{3}(d_{safe} + \Delta d_{phys})$. Whenever $\Delta d_{phys} > d_{jm}^{max}/\sqrt{3} - d_{safe}$, the passage undergoes*

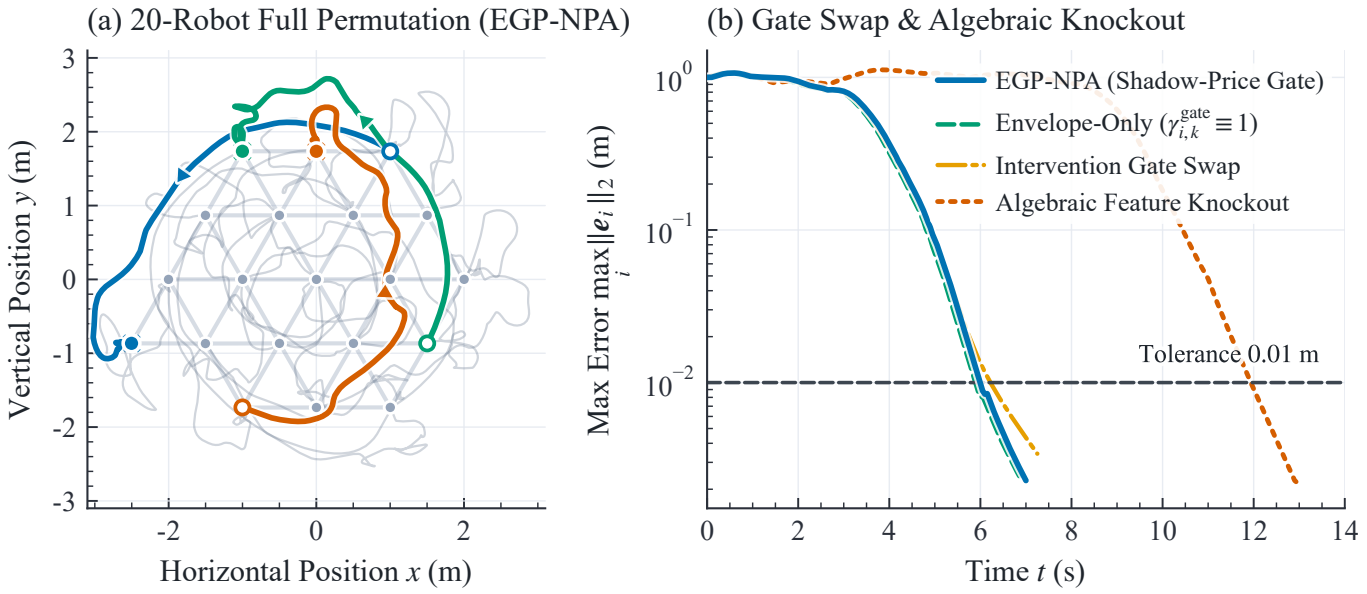


Fig. 5. Twenty-robot full-permutation trajectories and fixed-weight gate replacement and feature knockout responses. (a) Spatial trajectories of EGP-NPA on $N = 20$ random full permutation (slate lines: background fleet; colored bold curves: three large-span crossing robots); (b) maximum error convergence under fixed policy weights $\theta^\star$ comparing shadow-price gate $\gamma^{\text{gate},\mu}$, envelope-only gate $\gamma^{\text{gate}} \equiv 1$, intervention gate swap $\gamma^{\text{gate,int}}$, and algebraic feature knockout.

TABLE V
FULL-STACK MuJoCo SECOND-ORDER CHASSIS VERIFICATION COMBINING EGP-NPA, ONLINE DIVISION-FREE SPECTRAL ESTIMATION, AND CLOSED-NEIGHBORHOOD NPA ACROSS TWO PHYSICAL COMPENSATION REGIMES ($d_{\text{safe}} = 1.000$ m; 0 CONNECTIVITY VIOLATIONS IN ALL 64 RUNS)

Regime	Scale	System Configuration	Arrival (Mean)	$\min_t d_{ij}$ (m)	Viol. Steps	$\min_t \lambda_2$	$\hat{\lambda}_2$ RMSE	Final Err. (m)
Regime A (Look-Ahead, $S = 1.018$ m)	$N = 8$	Myopic Attraction + Phys-NPA	0/4 (—)	0.9817	610.2	6.963	0.0273	1.394
		EGP-NPA + Static Split ($B = 0$)	1/4 (10.12 s)	0.9798	442.2	6.847	0.0318	1.353
		EGP-NPA + Ideal Single-Integrator NPA	0/4 (—)	0.9792	4 231.0	7.071	0.0268	1.653
		EGP-NPA + Online Est. + Phys-NPA (Full)	**3/4 (13.88 s)**	0.8990	1 683.5	6.670	0.0337	**0.629**
	$N = 12$	Myopic Attraction + Phys-NPA	0/4 (—)	0.9952	1 679.2	10.319	0.0411	1.972
		EGP-NPA + Static Split ($B = 0$)	0/4 (—)	0.9976	850.8	9.632	0.0417	1.881
		EGP-NPA + Ideal Single-Integrator NPA	0/4 (—)	0.9937	4 190.2	10.184	0.0412	2.003
		EGP-NPA + Online Est. + Phys-NPA (Full)	**1/4 (14.54 s)**	0.8988	2 777.0	9.431	0.0438	**1.314**
Regime B (Conservative, $S = 1.045$ m, $\Delta d = 0.028$ m)	$N = 8$	Myopic Attraction + Phys-NPA	0/4 (—)	1.0240	0.0	7.237	0.0255	1.680
		EGP-NPA + Static Split ($B = 0$)	0/4 (—)	1.0260	0.0	7.159	0.0256	1.677
		EGP-NPA + Ideal Single-Integrator NPA	0/4 (—)	0.8987	4 722.0	7.027	0.0266	1.588
		EGP-NPA + Online Est. + Phys-NPA (Full)	0/4 (—)	**0.9910**	1 133.2†	6.997	0.0257	1.659
	$N = 12$	Myopic Attraction + Phys-NPA	0/4 (—)	1.0256	0.0	10.370	0.0393	2.056
		EGP-NPA + Static Split ($B = 0$)	0/4 (—)	1.0255	0.0	10.151	0.0394	2.077
		EGP-NPA + Ideal Single-Integrator NPA	0/4 (—)	0.9809	4 783.2	10.048	0.0393	1.974
		EGP-NPA + Online Est. + Phys-NPA (Full)	0/4 (—)	**1.0096**	**0.0**	10.033	0.0394	2.065

Note: †Under Regime B ($N = 8$), the full-stack configuration achieves 0 collision violation steps on 3 of 4 tasks ($\min_t d_{ij} \geq 1.0068$ m); on one local swap task, post-filter slew-rate clipping ($\Delta u_{\max} = 0.22$ m/s) coupled with observation noise causes static settling at $d_{ij} = 0.9910$ m $> 2r_{\text{body}} = 0.90$ m after channel closure.

topological pinch-off in $\mathcal{S}^{\text{phys}}$, disconnecting the initial and target configurations into separate path-connected components of $\mathcal{S}^{\text{phys}}$.

Proof. Claim (1) follows from $\min_{a<b} d_{ab}(0) = S$. For Claim (2), at the symmetric midpoint, $d_{i_1 i_2} \geq d^{\text{phys}}_{\text{safe}}$ and $d_{i_1 j}, d_{i_1 m}, d_{i_2 j}, d_{i_2 m} \geq d^{\text{phys}}_{\text{safe}}$ imply $(d_{i_1 i_2}/2)^2 + (d_{jm}/2)^2 \geq (d^{\text{phys}}_{\text{safe}})^2 \implies d_{jm} \geq \sqrt{3}(d_{\text{safe}} + \Delta d_{\text{phys}})$. □

Table V and Fig. 6 demonstrate both seamless estimator–policy coupling ($\hat{\lambda}_2$ RMSE $0.0255 \sim 0.0438$ and 0 connectivity violations across all 64 runs) and the physical margin–passability trade-off predicted by Theorem 3: under Regime A ($S = 1.018$ m), full-stack EGP-NPA achieves 3/4 arrivals on

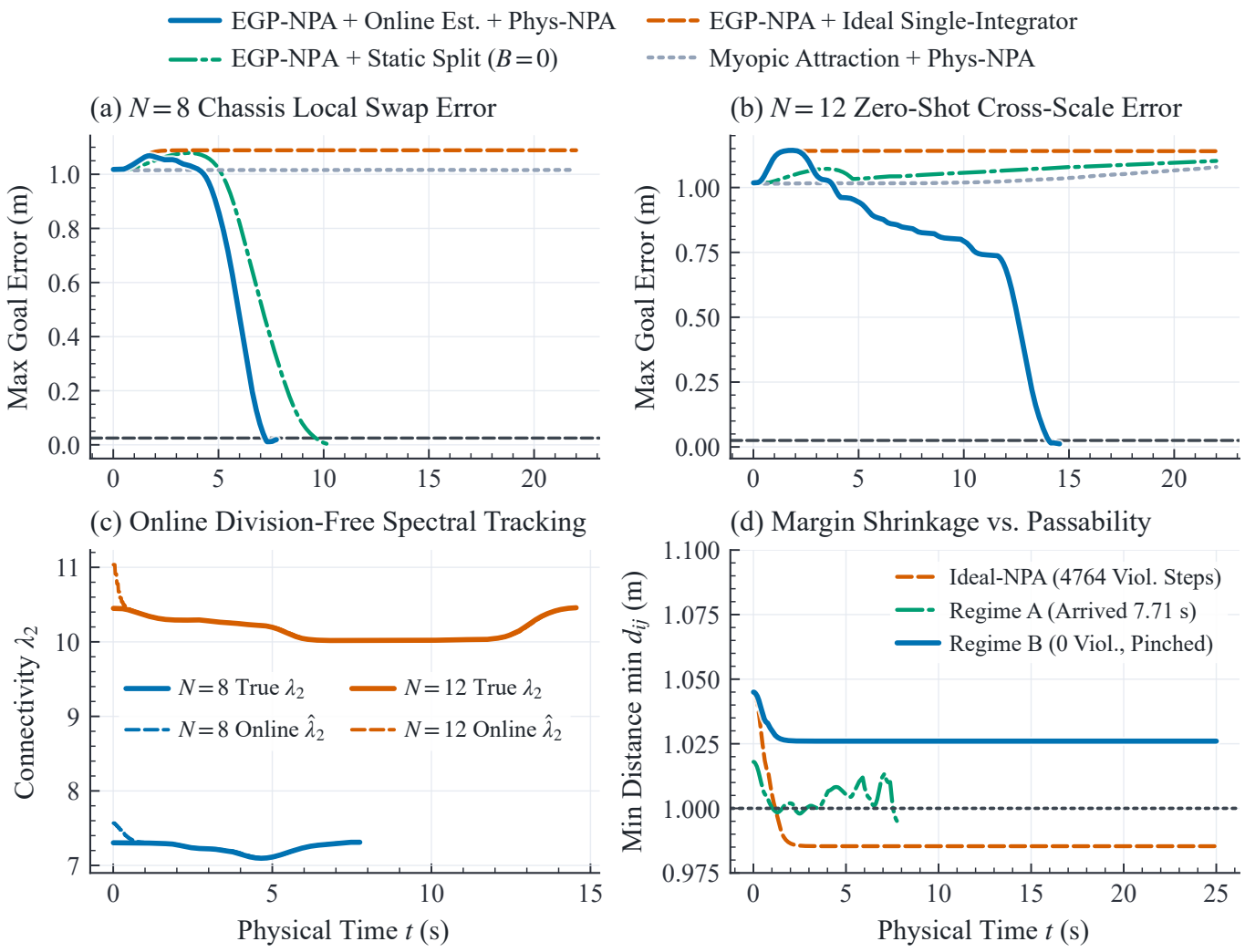


Fig. 6. Full-stack `MuJoCo` second-order omnidirectional chassis verification combining EGP-NPA, online division-free spectral estimation, and closed-neighborhood NPA. (a)–(b) Maximum goal error on $N = 8$ and $N = 12$ local swap tasks under Regime A; (c) real-time $\hat{\lambda}_2(t)$ tracking by the division-free spectral estimator; (d) trade-off between minimum inter-robot clearance $\min d_{ij}(t)$ and channel passability across Ideal-NPA, Regime A, and Regime B.

$N = 8$ and $1/4$ on $N = 12$ (vs. $0/8$ for Myopic and Ideal-NPA), whereas enforcing $\Delta d_{\text{phys}} = 0.028$ m under Regime B eliminates soft-boundary grazing (0 violations on $7/8$ tasks) at the cost of pinching off the three-body swap channels.

VI. Conclusion

This paper presented the Envelope Equivariant Graph Policy with Neighborhood Proximal Allocation (EGP-NPA) framework for cooperative deadlock resolution of multi-robot systems subject to mutually conflicting global connectivity and pairwise collision-avoidance scalar constraints. First, an exact rational second-order KKT certificate on a seven-robot symmetric contact configuration (Theorem 2) proved that safe and connected reachability does not imply reachability under myopic error-monotone safety filters, identifying lateral symmetry-breaking drive and dynamic quota-driven neighbor yielding as key physical mechanisms for deadlock resolution. Second, we developed an $\mathrm{SO}(2) \ltimes \mathbb{R}^2$ and permutation-equivariant GNN residual nominal policy (Proposition 1), proving why preserving $\mathrm{SO}(2)$ chiral orientation over full $\mathrm{O}(2)$ reflection equivariance is essential to prevent zero tangential output on symmetric head-on encounters. Third, we established anytime continuous-time safety invariance (Proposition 2), derived the closed-form solution for cooperative unfreezing of stationary neighbors via dynamic quota transfer (Proposition 3), and proved first-order $\mathcal{O}(\|e_k\|_2)$ residual suppression and goal equilibrium recovery under the stateless one-hop completion envelope $\rho_{i,k}$ (Proposition 4). Across 96 training-scale and zero-shot cross-scale benchmarks and 48 independent holdout instances, EGP-NPA achieved 100% safe arrival and zero safety violations (90.6% and 87.5% cross-scale arrival under matched half-budget two-seed retraining)—outperforming shield-consistent safe MARL ($53.1\%, p = 6.1 \times 10^{-5}$), static-split CBF shielded GNN ($3.1\%, p = 9.3 \times 10^{-10}$), task-adapted DGPPO ($0\%$ violation-free arrival), primal-dual MAPPO-Lagrangian (0% violation-free arrival), tangential circulation heuristics ($40.6\%, p = 3.8 \times 10^{-6}$), and myopic CBF filtering ($15.6\%, p = 1.5 \times 10^{-8}$)—while second-order `MuJoCo` simulations with online division-free spectral estimation validated full-stack in-

tegration and characterized the physical safety-margin versus channel-passability trade-off (Theorem 3). Future work will incorporate multi-scale boundary-clearance features and multi-seed curriculum training on larger lattices, as well as extensions to nonholonomic and field-of-view-constrained multi-robot teams.

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